

Unit 0 Science Practices:**Set A**

- ☐ Independent variables
- ☐ Confounding variables
- ☐ Dependent variables
- ☐ Random assignment
- ☐ Hypothesis
- ☐ Generalizability
- ☐ Quantitative data
- ☐ Qualitative data

Set B

- ☐ Institutional review
- ☐ Informed consent
- ☐ Informed assent
- ☐ Protection from harm
- ☐ Confidentiality
- ☐ Deception
- ☐ Debriefing

Set C

- ☐ Central tendency
- ☐ Mean
- ☐ Median
- ☐ Mode
- ☐ Variation
- ☐ Percentile rank
- ☐ Range

Set D

- ☐ Normal curve
- ☐ Bimodal distribution
- ☐ Standard deviation
- ☐ Regression toward the mean
- ☐ Positive skew
- ☐ Negative skew
- ☐ Scatterplot
- ☐ Correlation
- ☐ Positive Correlation
- ☐ Negative Correlation
- ☐ Correlation coefficient

Set E

- ☐ Population
- ☐ Sample
- ☐ Representative sample
- ☐ Random sampling
- ☐ Convenience sampling
- ☐ Representation of participants
- ☐ Experimental group
- ☐ Control group
- ☐ Single-blind study
- ☐ Double-blind study

Set F

- ☐ Operational definition
- ☐ Case study
- ☐ Meta-analysis
- ☐ Naturalistic observation
- ☐ Structured interviews
- ☐ Likert scale
- ☐ Effect size
- ☐ Statistical significance
- ☐ Replication
- ☐ Peer review

Set G

- ☐ Falsifiability (pertaining to hypotheses)
- ☐ Placebo
- ☐ Placebo effect
- ☐ Overconfidence
- ☐ Directionality problem (in correlation)
- ☐ Third variable problem (in correlation)
- ☐ Experimenter bias
- ☐ Social desirability bias
- ☐ Cultural norms
- ☐ Confirmation bias
- ☐ Hindsight bias

Set A: Research Variables & Logic

- **Independent variables**

- The factor *manipulated* by the researcher to determine its effect.
- 研究咖啡因对记忆的影响，**咖啡因摄入量**就是自变量

- **Dependent variables**

- The outcome factor that is *measured*; the variable that may change in response to manipulations.
- 在上述研究中，**记忆力测试的分数**就是因变量

- **Confounding variables**

- Factors other than the independent variable that might produce an effect in an experiment.
- 参与者的**睡眠质量**可能会干扰对咖啡因效果的判断

- **Random assignment**

- Assigning participants to experimental and control groups by chance.
- 通过**抽签**决定谁喝真咖啡，谁喝无咖啡因饮料
- 判断是否是 experiment 的 key point

- **Hypothesis**

- A testable prediction, often implied by a theory.
- 假设：**摄入咖啡因的学生在记忆测试中表现更好**

- **Generalizability**

- The extent to which research findings can be applied to the broader population.
- 研究结果是否能从这 10 名学生**推及**到全校学生

- **Quantitative data**

- Information that can be measured and written down with numbers.
- 记录测试得到的**具体分数**

- **Qualitative data**

- Descriptive data used to understand concepts, thoughts, or experiences.
- 记录参与者对测试过程的**主观感受描述**

Set B: Ethical Guidelines

- **Institutional review**

- A committee that reviews research proposals for ethical compliance.
- 实验开始前必须提交给 IRB 审核

- **Informed consent**

- Giving potential participants enough information about a study to enable them to choose whether they wish to participate.
- 实验前让学生了解流程并签署同意书

- **Informed assent**

- Agreement to participate from a minor or someone unable to give legal consent
- 针对未成年人，除了家长同意，还需要孩子本人表达参与意愿
- David，不要再拼错了

- **Protection from harm**

- Ensuring participants do not endure any unnecessary physical or psychological distress.
- 确保实验不会对学生造成持久的心理压力

- **Confidentiality**

- Keeping all information about participants private and anonymous.
- 发布数据时隐去学生姓名，只用编号代替，只看数据没办法知道是谁的

- **Deception**

- Intentionally misleading participants about the true nature of the experiment.
- 告诉学生是在测视力，实际在测从众行为
- Proper deception is acceptable，但无意义的 deception 会被 IRB 驳回

- **Debriefing**

- The post-experimental explanation of a study, including its purpose and any deceptions.
- 实验结束后，老师真相大白，解释实验目的和流程，if 有 deception，需要解释原因
- 无论有没有 deception，都需要有 debriefing 的环节

Set C & D: Statistics

- **Central tendency**

- A single score that represents a whole set of scores.
- 使用一个数值来概括全班的整体表现
- 包括 3M

- **Mean**

- The arithmetic average of a distribution.
- 把全班分数加起来除以总人数

- **Median**

- The middle score in a distribution.
- 将全班分数从低到高排列，找到最中间那个数

- **Mode**

- The most frequently occurring score(s) in a distribution.
- 班里出现次数最多的分数，比如有 10 个人都考了 85 分
- 如果都只出现了一次，则本 data set 没有 mode

- **Variation**

- How similar or diverse the scores are.
- 观察班级内学生分数的差异程度

- **Percentile rank**

- The percentage of scores in a distribution that are lower than or equal to a specific score.
- 你的分数超过了全校 90% 的人

- **Range**

- The difference between the highest and lowest scores in a distribution.
- 最高分 98 减去最低分 40，range 是 58

- **Normal curve**

- A symmetrical, bell-shaped curve that describes the distribution of many types of data.
- 大多数人的身高都集中在平均值附近

- 只要 sample size 够大，很多 characteristics 都会是 normal distribution

- **Bimodal distribution**

- A distribution with two clear peaks.
- 一天车流量明显有两个高峰，早高峰和晚高峰

- **Standard deviation**

- A computed measure of how much scores vary around the mean score.
- 反映分数是紧贴平均分还是离散得很远

- **Regression toward the mean**

- The tendency for extreme scores to fall back toward their average.
- 第一次考了破纪录高分的球星，下次表现往往会回归平庸

- **Positive skew**

- A distribution where the tail points toward the higher values (right).
- 题目太难导致大部分人不及格，只有极少数人考了高分
- Mean > Median

- **Negative skew**

- A distribution where the tail points toward the lower values (left).
- 题目太简单导致大部分人都拿高分，只有极少数人不及格
- Mean < Median

- **Scatterplot**

- A graphed cluster of dots representing the values of two variables.
- 在图表上画出身高与鞋码对应的数据点
- Correlational study 画出 data 的方式

- **Correlation**

- A measure of the extent to which two factors vary together.
- 研究观看暴力电视与攻击性行为之间的联系
- Moses 上了 AP Psych 就被斯坦福录了，所以能得出什么结论呢？🐶
- Correlation 中不分 IV / DV

- **Positive correlation**

- Both variables increase or decrease together.
- 复习时间越多，考试成绩越高
- **Negative correlation**
 - As one variable increases, the other decreases.
 - 缺勤次数越多，考试成绩越低
 - 两个 variables 的变化方向不一致
- **Correlation coefficient**
 - A statistical index of the relationship between two things (from -1.00 to +1.00).
 - $r = +0.85$ ，说明两者有很强的正相关
 - AAQ Question C 分析 r 时，需要说明 direction + strength

Set E: Sampling & Experimental Design

- **Population**
 - All those in a group being studied, from which samples may be drawn.
 - 研究的对象是全中国所有的高中生
 - 所有的 potential samples
- **Sample**
 - A subset of the population used for the actual study.
 - 实际参与实验的 500 名学生
- **Representative sample**
 - A sample that accurately reflects the characteristics of the population.
 - 样本中男女比例、地区分布要与全国情况一致
 - 有代表性的 sample 才能 generalize to larger population
 - Demographic factors 要多样
- **Random sampling**
 - A sample that fairly represents a population because each member has an equal chance of inclusion.
 - 通过电脑随机生成的名单来抽取受试者

- 和 random assignment 的区别不想再说，别看错了球球了
- **Convenience sampling**
 - Choosing individuals who are easiest to reach.
 - 为了省事，只调查了 **Psych** 班里的学生
- **Representation of participants**
 - The degree to which the sample matches the population.
 - 这 500 名受试者是否能代表全体高中生？需要判断一下 degree of generalizability
- **Experimental group**
 - The group exposed to the treatment (independent variable).
 - 在实验中服用新开发药物的那组人
- **Control group**
 - The group not exposed to the treatment, serving as a comparison.
 - 在实验中服用安慰剂的那组人
 - 吃 placebo 的
- **Single-blind study**
 - Participants do not know whether they are in the experimental or control group.
 - 病人不知道自己吃的是真药还是小糖片
- **Double-blind study**
 - Neither the participants nor the researchers know who is in which group.
 - 学生和代课老师都不知道哪组是实验组
 - Avoid experimenter bias （你现在有什么感觉吗？真的没有吗~~ 被试：？）

Set F: Research Methods & Methodology

- **Operational definition**
 - A carefully worded statement of the exact procedures used in a research study.
 - 将学习压力定义为唾液中 **cortisol** 的含量
 - 让读者 or 后续的研究人员知道你到底是怎么测量 variables 的

- 记得找 questionnaire 或者是任何的 quantifiable measurement! 不过在 AAQ Question B, 是让你找文章中的 OD, 不是你凭空想一个

- **Case study**

- A descriptive technique in which one individual or group is studied in depth.
- 对一名大脑受损的罕见病例进行长达十年的追踪研究
- Rare / unethical to duplicate
- H.M. for anterograde amnesia, Broca's / Wernicke's aphasia, prefrontal cortex study

- **Meta-analysis**

- A statistical procedure for combining and analyzing data from many different studies.
- 综合过去 20 年里所有关于睡眠与记忆的研究得出结论

- **Naturalistic observation**

- Observing and recording behavior in naturally occurring situations without manipulation.
- 在操场上偷偷观察并记录小朋友玩耍时的社交互动
- 睁大你的眼睛看看怎么拼的哈, 不要再给我写什么 natural study 了

- **Structured interviews**

- A research procedure in which all participants are asked the same set of questions.
- 对所有应聘者统一询问三个固定问题
- 别自由发挥, 不让 confounding variables 影响结果

- **Likert scale**

- A survey scale used to measure attitudes or opinions (e.g., 1-5 scale).
- 请对 Irene 的 AP Psych 满意度打分, 1 分是非常不满, 5 分是非常满意

- **Effect size**

- The strength of the relationship between two variables.
- 虽然结果显著, 但这种药对减肥的实际效果大小只有 0.5 斤
- Pearson's r 是一种 effect size

- **Statistical significance**

- A statistical statement of how likely it is that an obtained result occurred by chance.

- p 值小于 0.05（不需要记，只是写到这方便理解）说明实验结果不是乱撞上的，不是 by chance 而是真实的情况

- **Replication**

- Repeating the essence of a research study with different participants.
- 在中国做完实验后，去美国用同样的步骤再做一次
- Replication 的前提是有非常清楚的 OD

- **Peer review**

- Evaluation of scientific work by others working in the same field.
- 论文发表前由其他心理学专家进行盲审和挑错

Set G: Cognitive Biases & Research Problems

- **Falsifiability**

- The possibility that a hypothesis can be proven false.
- 科学假设必须是可被证伪的，比如：我们坚信小栗小蛋糕是最可爱的，这个导致了你考 5 分，这个就没办法证伪 or 证实，所以这个不是科学
- Psychoanalysis / psychodynamics 的许多理论就因为缺乏 falsifiability 而受到批评

- **Placebo**

- An inert substance or condition given to the control group.
- 给 control group 吃外表一模一样的淀粉片

- **Placebo effect**

- Experimental results caused by expectations alone.
- 病人以为吃了药，结果心理作用导致病情真的好转了
- Single blind 就可以做到避免这个，不过实际能做到 double blind 更好

- **Overconfidence**

- The tendency to be more confident than correct.
- 学生觉得自己复习得很好能考 5 分，结果实际只考了 3 分

- **Directionality problem**

- When a correlation exists, it's unclear which variable caused the change in the other.

- 是焦虑导致了失眠，还是失眠导致了焦虑？

- **Third variable problem**

- An external variable that leads to a mistaken correlation between two factors.
- 夏天冰淇淋销量和溺水人数正相关，其实是因为气温升高这个 variable

- **Experimenter bias**

- When a researcher's expectations influence the results.
- 老师潜意识里希望实验成功，从而不自觉地鼓励了实验组

- **Social desirability bias**

- Giving answers that make one look good in the eyes of others.
- 在调查问卷中谎称自己从不撒谎

- **Cultural norms**

- The rules and expectations of a society.
- 中国人过年给红包是一种 norm，来都来了大过年的给孩子的
- Display rules 是 cultural norms 在 emotion expression 的体现

- **Confirmation bias**

- A tendency to search for information that supports our preconceptions.
- 只关注能证明玩游戏有害的新闻，无视能证明有益的研究

- **Hindsight bias**

- The tendency to believe, after learning an outcome, that one would have foreseen it.
- 考试答案公布是 C，你拍大腿说：我当时就说选 C 吧 :)

Unit 1: Biological Bases of Behavior

1.1 - Interaction of Heredity and Environment

- | | | |
|---|---|---|
| ☐ Heredity | ☐ Evolutionary perspective | ☐ Family studies |
| ☐ Nature | ☐ Natural selection | ☐ Adoption studies |
| ☐ Nurture | ☐ Eugenics | |
| ☐ Genetic predisposition | ☐ Twin studies | |

1.2 - Overview of the Nervous System

- | | | |
|--|---|---|
| ☐ Central nervous system
(brain and spinal cord) | ☐ Autonomic nervous system | ☐ Parasympathetic nervous system |
| ☐ Peripheral nervous system | ☐ Sympathetic nervous system | ☐ Somatic nervous system |

1.3 - The Neuron and Neural Firing

- | | | |
|---|---|--|
| ☐ Neurons | ☐ Excitatory neurotransmitters | ☐ Psychoactive drugs |
| ☐ Glial cells | ☐ Inhibitory neurotransmitters | ☐ Agonists |
| ☐ Reflex arc | ☐ Serotonin | ☐ Antagonists |
| ☐ Sensory neurons | ☐ Norepinephrine | ☐ Reuptake inhibitors |
| ☐ Motor neurons | ☐ Glutamate | ☐ Stimulants |
| ☐ Interneurons | ☐ GABA | ☐ Caffeine |
| ☐ Neural transmission | ☐ Endorphins | ☐ Cocaine |
| ☐ Action potential | ☐ Substance p | ☐ Depressants |
| ☐ All-or-nothing principle | ☐ Acetylcholine | ☐ Alcohol |
| ☐ Depolarization | ☐ Hormones | ☐ Hallucinogens |
| ☐ Refractory period | ☐ Adrenaline | ☐ Marijuana |
| ☐ Resting potential | ☐ Leptin | ☐ Opioids |
| ☐ Reuptake | ☐ Ghrelin | ☐ Heroin |
| ☐ Threshold | ☐ Melatonin | ☐ Tolerance |
| ☐ Multiple sclerosis | ☐ Oxytocin | ☐ Addiction |
| ☐ Myasthenia gravis | | ☐ Withdrawal |
| ☐ Dopamine | | |

1.4 - The Brain

- | | | |
|---|--|--|
| ☐ Brain stem | ☐ Corpus callosum | ☐ Hemispheric specialization |
| ☐ Medulla | ☐ Occipital lobes | ☐ Broca's area |
| ☐ Reticular activating system
(RAS) | ☐ Temporal lobes | ☐ Wernicke's area |
| ☐ Reward center | ☐ Parietal lobes | ☐ Aphasia (Broca's and
Wernicke's) |
| ☐ Cerebellum | ☐ Association areas | ☐ Contralateral hemispheric
organization |
| ☐ Cerebral cortex | ☐ Somatosensory cortex | ☐ Plasticity |
| ☐ Limbic system | ☐ Frontal lobes | ☐ EEG |
| ☐ Thalamus | ☐ Linguistic processing | ☐ fMRI |
| ☐ Hypothalamus | ☐ Higher-order thinking | ☐ Lesioning |
| ☐ Pituitary gland | ☐ Executive functioning | |
| ☐ Hippocampus | ☐ Prefrontal cortex | |
| ☐ Amygdala | ☐ Motor cortex | |
| | ☐ Split brain research | |

1.5 - Sleep

- ☐ Consciousness
- ☐ Circadian rhythm (sleep/wake cycle)
- ☐ Jet lag
- ☐ Shift work
- ☐ NREM stage 1
- ☐ Hypnagogic sensations
- ☐ NREM stage 2
- ☐ NREM stage 3
- ☐ REM sleep
- ☐ REM rebound
- ☐ Activation-synthesis (dreams)
- ☐ Consolidation theory (dreams)
- ☐ Restoration of resources
- ☐ Insomnia
- ☐ Narcolepsy
- ☐ REM sleep behavior disorder
- ☐ Sleep apnea
- ☐ Somnambulism

1.6 - Sensation

- ☐ Sensation
- ☐ Transduction
- ☐ Absolute threshold
- ☐ Just-noticeable difference
- ☐ Sensory adaptation
- ☐ Weber's law
- ☐ Sensory interaction
- ☐ Synesthesia
- ☐ Retina
- ☐ Blind spot
- ☐ Visual nerve
- ☐ Lens
- ☐ Accommodation
- ☐ Nearsightedness
- ☐ Farsightedness
- ☐ Photoreceptors
- ☐ Rods
- ☐ Cones (blue, green, red)
- ☐ Trichromatic theory
- ☐ Opponent-process theory
- ☐ Afterimages
- ☐ Ganglion cells
- ☐ Dichromatism
- ☐ Monochromatism
- ☐ Prosopagnosia
- ☐ Blindsight
- ☐ Wavelength
- ☐ Pitch
- ☐ Amplitude
- ☐ Loudness
- ☐ Pitch perception
- ☐ Place theory
- ☐ Volley theory
- ☐ Frequency theory
- ☐ Sound localization
- ☐ Conduction deafness
- ☐ Sensorineural deafness
- ☐ Olfactory system
- ☐ Thalamus
- ☐ Pheromones
- ☐ Gustation
- ☐ Sweet, sour, salty, bitter, umami, oleogustus
- ☐ Taste receptors
- ☐ Supertasters
- ☐ Medium tasters
- ☐ Nontasters
- ☐ Warm/cold receptors
- ☐ Gate control theory
- ☐ Phantom limb
- ☐ Vestibular sense
- ☐ Semicircular canals
- ☐ Kinesthesia

1.1 - Interaction of Heredity and Environment

- **Heredity**
 - The genetic transfer of characteristics from parents to offspring.
 - 孩子继承了父母的高个子基因
- **Nature**
 - The influence of our inherited characteristics on our personality and physical growth.
 - 一个人天生就比较内向
 - 有一些看到就应该定位是 bio/nature 的词: temperament / diathesis / mature ... (还有啥?)
- **Nurture**
 - The influence of environmental factors on a person.
 - 后天培养和家庭环境影响了孩子的成就
- **Genetic predisposition**
 - An increased likelihood of developing a particular disease based on a person's genetic makeup.
 - 家族中有抑郁症史, 个体更有可能表现出抑郁倾向
- **Evolutionary perspective**
 - Study of the evolution of behavior and the mind, using principles of natural selection.
 - 从进化角度解释为什么人类普遍害怕蛇和蜘蛛
 - Survival & reproduction
- **Natural selection**
 - Traits leading to increased reproduction and survival will be passed on to succeeding generations.
 - 跑得快的远古人类更有可能生存并传下基因
- **Eugenics**
 - The study of agency under social control that may improve or impair the racial qualities of future generations.
 - 历史上通过筛选所谓优良基因来改善人口质量的极端主张
 - Social Darwinism → 是不对的!

- Eu- “good” 记得 eustress 吗？

- **Twin studies**

- Research comparing identical twins and fraternal twins to determine the relative importance of heredity and environment.
- 观察分开抚养的双胞胎在智力上的相似性
- Longitudinal study / correlation / case study, 因为 ethics 很难真的做 experiment

- **Family studies**

- Research examining the tendency of traits to run in families.
- 研究近亲之间患某种精神疾病的概率

- **Adoption studies**

- Research comparing adopted children to their biological and adoptive parents.
- 发现孩子的性格更像亲生父母而非养父母

1.2 - Overview of the Nervous System

- **Central nervous system**

- Composed of the brain and spinal cord.
- 作为身体的指挥中心处理所有信息
- 不会还有同学记不住 nervous system 的分类吧！不会吧！

- **Peripheral nervous system**

- The sensory and motor neurons that connect the CNS to the rest of the body.
- 将手指触碰火苗的感觉传回神经中枢

- **Autonomic nervous system**

- Part of the PNS that controls the glands and muscles of internal organs.
- 自动调节心跳和消化，不需要大脑刻意去想
- 包括 sympathetic / parasympathetic

- **Sympathetic nervous system**

- Arouses the body, mobilizing its energy in stressful situations.
- 遇到老虎时，它让你瞳孔放大、心跳加快、胃部不工作
- Arousal → Hormones: adrenaline / cortisol

- **Parasympathetic nervous system**

- Calms the body, conserving its energy.
- 危机过后，它让身体恢复平静和节能状态
- Calm and digest 考完 SAT 才感觉有点饿了

- **Somatic nervous system**

- Controls the body's skeletal muscles.
- 你想挥手打招呼，它控制手臂肌肉移动
- Voluntary movement
- 和 soma- 相关，就是身体的意思：somatosensory cortex (tactile sense); soma of neuron

1.3 - The Neuron and Neural Firing

- **Neurons**

- Basic building blocks of the nervous system.
- 大脑中负责传递信息的神经细胞

- **Glial cells**

- Cells in the nervous system that support, nourish, and protect neurons.
- 为神经元提供营养的后勤细胞

- **Reflex arc**

- A neural pathway that controls a reflex.
- 手碰到热锅立刻缩回来，信息没到大脑就在脊髓完成了回路
- 和 developmental psych 的那个 rooting reflex 不一样，rooting 只有婴儿有，但是 reflex arc 是啥时候都存在，理论上路易十六也会有（目移）

- **Sensory neurons**

- Carry incoming information from the sensory receptors to the CNS.
- 把眼睛看到的光信号传给大脑

- **Motor neurons**

- Carry outgoing information from the CNS to muscles and glands.
- 大脑发出指令让腿部肌肉跑步

- **Interneurons**

- Neurons within the CNS that communicate internally and intervene between sensory and motor outputs.
- 在脊髓中连接感觉神经元和运动神经元的中间人

- **Neural transmission**

- The process by which signaling molecules are released by a neuron and bind to and activate the receptors of another neuron.
- 电信号在神经元之间转化和传递的过程

- **Action potential**

- A brief electrical charge that travels down an axon.
- 神经冲动像闪电一样沿轴突传导

- **All-or-nothing principle**

- A neuron's reaction of either firing or not firing.
- 就像开枪，扣动扳机要么发射，要么不发射，没有半发子弹

- **Depolarization**

- The process during the action potential when sodium is rushing into the cell.
- 细胞膜内外电荷反转，由负变正
- 在做题的时候，depolarization 就意味着 action potential / neural impulse

- **Refractory period**

- A period of inactivity after a neuron has fired.
- 神经元发射后的短暂冷却期，此时无法再次发射
- 分成 absolute / relative (但是确实没见过题考过这么细)

- **Resting potential**

- The positive-outside/negative-inside state of a neuron's membrane.
- 神经元在没干活时的静息状态

- **Reuptake**

- A neurotransmitter's reabsorption by the sending neuron.

- 神经递质完成任务后被**重新回收利用**
- SSRI → MDD
- **Threshold**
 - The level of stimulation required to trigger a neural impulse.
 - 刺激必须达到一定**强度**才能让神经元放电
- **Multiple sclerosis**
 - A disease in which the immune system eats away at the protective covering of nerves.
 - 髓鞘受损导致**神经信号传递变慢**或中断
 - MS (myelin sheath) → MS
- **Myasthenia gravis**
 - A chronic autoimmune disease that affects the neuromuscular junction and produces serious weakness of voluntary muscles.
 - 表现为眼睑下垂或严重的**肌肉无力**
 - ACh
- **Dopamine**
 - Neurotransmitter associated with movement, learning, attention, and emotion.
 - 中彩票时的**奖赏感**和愉悦感
 - 和 addiction 有关，比如说上瘾玩游戏，是因为 dopamine 的奖励机制
 - (+) schizophrenia (只是 correlation) (-) Parkinson's disease
- **Excitatory neurotransmitters**
 - Increase the likelihood that the neuron will fire an action potential.
 - 给神经递质发出的**加油**指令，速速传递消息！
- **Inhibitory neurotransmitters**
 - Decrease the likelihood that the neuron will fire an action potential.
 - 给神经递质发出的**刹车**指令 
- **Serotonin**
 - Affects mood, hunger, sleep, and arousal.

- 血清素不足可能导致**抑郁情绪**
- 所以 SSRI 的本质是通过抑制 serotonin 的 reuptake → 更多 serotonin 传递信息
- **Norepinephrine**
 - Helps control alertness and arousal.
 - 在压力下保持**警觉和兴奋**
- **Glutamate**
 - A major excitatory neurotransmitter involved in memory.
 - 帮助你更好地**记住单词**
- **GABA**
 - A major inhibitory neurotransmitter.
 - 帮助大脑**冷静下来**，减少焦虑
 - Alcohol 是 GABA agonist → 让你 calm down → impulse control ↓ → 会放松 / 嗜睡 / 反应变慢（alcohol 作用的主要地点是 prefrontal cortex + cerebellum）
- **Endorphins**
 - Natural, opiate-like neurotransmitters linked to pain control and pleasure.
 - 长跑后的**快感**，减轻身体的疼痛
 - Painkiller → 所以可以说是一种 negative reinforcement
- **Substance P**
 - A neurotransmitter involved in pain perception.
 - 手指扎破时向大脑发送**疼痛信号**
- **Acetylcholine (ACh)**
 - Enables muscle action, learning, and memory.
 - 控制肌肉收缩，缺失与**阿尔兹海默症**有关
 - MG 肌无力！
- **Hormones**
 - Chemical messengers that are manufactured by the endocrine glands.
 - 通过**血液**运输的长效化学信息

- 本质和 neurotransmitter 是一样的，只是作用的地点不一致
- **Adrenaline**
 - Hormone secreted in response to stress, increasing rates of blood circulation and breathing.
 - 被 Irene 提问的时候你心跳狂飙
- **Leptin**
 - Hormone that helps regulate energy balance by inhibiting hunger.
 - 告诉大脑你吃饱了 → 传递 full signal
 - 这俩和 satiety 相关的 hormones 在 UNIT 4 Motivation~
- **Ghrelin**
 - Hormone secreted by empty stomach that sends "I'm hungry" signals to the brain.
 - 肚子饿得咕咕叫时，它催你快去吃饭 "咕"relin
- **Melatonin**
 - Hormone that helps regulate sleep-wake cycles.
 - 天黑时分泌增多，让你产生睡意
 - 和光线有关：睡前看手机，蓝光太多大脑就不会分泌 melatonin
- **Oxytocin**
 - Hormone associated with social bonding and trust.
 - 拥抱时分泌的爱与信任的荷尔蒙
- **Psychoactive drugs**
 - Chemicals that affect the central nervous system and alter activity in the brain.
 - 改变情绪和认知的化学物质
- **Agonists**
 - Drugs that increase the action of a neurotransmitter.
 - 模仿神经递质的作用 Mimicry & activation
- **Antagonists**
 - Drugs that block the function of a neurotransmitter.
 - 阻断受体，让神经递质无法工作 Block & inhibition
- **Reuptake inhibitors**

- Drugs that prevent the reabsorption of neurotransmitters.
- 让神经递质在间隙停留更久，从而增强效果
- Again! SSRI
- **Stimulants**
 - Drugs that excite neural activity and speed up body functions.
 - 让你精神百倍的物质
 - 咖啡！上课喝咖啡，那考试前也喝，想起来的知识点可能更多，这叫什么？—— State dependent memory (Unit 2)
- **Caffeine**
 - A mild stimulant found in coffee and tea.
 - 早起喝一杯咖啡来提神
- **Cocaine**
 - A powerful and addictive stimulant.
 - 导致极度兴奋并具有极强成瘾性
 - Indirect agonist → Cocaine inhibits the reuptake of dopamine
- **Depressants**
 - Drugs that reduce neural activity and slow body functions. 让神经系统变迟钝
- **Alcohol**
 - The most common depressant.
 - 喝醉后反应迟钝、走路不稳
- **Hallucinogens**
 - Psychedelic drugs that distort perceptions and evoke sensory images in the absence of sensory input.
 - 看到并不存在的幻象
 - 所以看到 halluci- 就是和 distorted perception 相关
- **Marijuana**
 - A drug that causes mild hallucinations and relaxation.

- 吸食后感到放松且感知力改变

- **Opioids**

- Drugs used to treat pain but are highly addictive.
- 强效止痛药，如吗啡

- **Heroin**

- An extremely addictive opioid drug.
- 极难戒掉的毒品

- **Tolerance**

- Diminishing effect with regular use of the same dose of a drug.
- 以前喝一杯瑞幸就清醒，现在要喝三杯才有同样的效果

- **Addiction**

- Compulsive drug craving and use, despite adverse consequences.
- 无法停止吸烟，即便知道伤害身体

- **Withdrawal**

- Discomfort and distress that follow discontinuing an addictive drug.
- 戒烟后的焦虑和渴望感
-

1.4 - The Brain

- **Brain stem**

- The oldest part and central core of the brain, responsible for automatic survival functions.
- 维持生命的底线 Life-sustaining functions

- **Medulla**

- Base of the brainstem; controls heartbeat and breathing.
- 如果这里受损，人就无法呼吸 or 心跳

- **Reticular activating system (RAS)**

- Nerve network in the brainstem that plays an important role in controlling arousal.
- 负责你的觉醒和清醒状态

- **Reward center**

- Region of the brain that produces feelings of pleasure.
- 大脑中的 **dopamine 通路**，让人想重复愉悦行为
- **Cerebellum**
 - The “little brain” at the rear of the brainstem; coordinates voluntary movement and balance.
 - 让你能平稳地走平衡木 Balance & coordination & posture
- **Cerebral cortex**
 - The intricate fabric of interconnected neural cells covering the cerebral hemispheres.
 - 大脑表层那层褶皱，负责高级认知
- **Limbic system**
 - Neural system associated with emotions and drives.
 - 大脑的情感中心
- **Thalamus**
 - The brain’s sensory control center.
 - 除了 **olfaction**，所有的感觉信息都要经过这个 **relay station**，再到各个感觉处理皮层
- **Hypothalamus**
 - Directs eating, drinking, body temperature; helps govern the endocrine system.
 - 维持身体的内环境稳态 homeostasis
- **Pituitary gland**
 - The endocrine system’s most influential gland.
 - 主宰其他腺体的总司令 master gland
- **Hippocampus**
 - A neural center located in the limbic system that helps process explicit memories for storage.
 - 如果这里坏了，你就记不住 **psych** 的知识点辣！
 - Memory formation & navigation
 - H.M. anterograde amnesia 也是因为 hippocampus 坏了（但还有 implicit memory）
- **Amygdala**
 - Two lima-bean-sized neural clusters in the limbic system; linked to emotion, especially fear and aggression.

- 看到蛇时产生的恐惧反应 “fear center”
- **Corpus callosum**
 - The large band of neural fibers connecting the two brain hemispheres.
 - 左右脑沟通的桥梁
 - Epilepsy → 之前的治疗方式是切 CC → split brain → 知道了 brain lateralization
- **Occipital lobes**
 - Portion of the cerebral cortex lying at the back of the head; includes areas that receive information from the visual fields.
 - 如果你后脑勺挨一记闷棍，可能会眼冒金星
 - （0.0）像不像你的眼睛 ∴ o lobe → vision
- **Temporal lobes**
 - Portion of the cerebral cortex lying roughly above the ears; includes the auditory areas.
 - 负责处理听到的声音
- **Parietal lobes**
 - Portion of the cerebral cortex lying at the top of the head; receives sensory input for touch and body position.
 - 感知冷热和触碰的部位 somatosensory cortex 在这里
- **Association areas**
 - Areas of the cerebral cortex that are involved in higher mental functions such as learning, remembering, thinking, and speaking.
 - 大脑中不直接负责感官或运动，而是负责深度加工的区域
- **Somatosensory cortex**
 - Area at the front of the parietal lobes that registers and processes body touch and movement sensations.
 - Leo 脚趾被 Jason 踩了，大脑在这个区域产生痛觉感应
- **Frontal lobes**
 - Involved in speaking and muscle movements and in making plans and judgments.

- 大脑的**理智中心**
- **Linguistic processing**
 - The process of understanding and using language.
 - 大脑对文字和语音的**解析** 左脑！
- **Higher-order thinking**
 - Complex cognitive processes including problem solving and decision making.
 - 思考量子力学等**抽象问题**
- **Executive functioning**
 - High-level cognitive processes that help individuals manage goals and control behaviors.
 - 在诱惑面前**控制住**自己不去玩手机
- **Prefrontal cortex**
 - Responsible for thinking, planning, and language.
 - 前额叶受损可能导致性格大变，变得**冲动易怒**
- **Motor cortex**
 - An area at the rear of the frontal lobes that controls voluntary movements.
 - 大脑发令让**手指打字写 FRQ** 的源头
- **Split brain research**
 - Studies on people whose corpus callosum has been severed.
 - 揭示了**左右脑分工**的秘密
- **Hemispheric specialization**
 - The idea that certain cognitive functions are localized in one hemisphere or the other.
 - 左脑侧重**逻辑和语言**，右脑侧重**空间感和 art**
- **Broca's area**
 - Controls language expression; an area in the frontal lobe that directs the muscle movements involved in speech.
 - 受损后你能听懂话，但**无法流利表达**
- **Wernicke's area**
 - Controls language reception; an area in the temporal lobe involved in language comprehension.

- 受损后你说话很顺溜但**满嘴胡言**，且听不懂别人的话

- **Aphasia**

- Impairment of language, usually caused by left hemisphere damage either to Broca's area or to Wernicke's area.
- 脑卒中后出现的**言语障碍** 有以上两个脑区的 aphasia

- **Contralateral hemispheric organization**

- The left hemisphere controls the right side of the body, and vice versa.
- 右手被扎时，**左脑** process 疼的新号

- **Plasticity**

- The brain's ability to change, especially during childhood, by reorganizing after damage or by building new pathways based on experience.
- 盲人的听觉区域可能会**接管**部分视觉区域的功能
- 越年轻越有 plasticity (或者有的时候就叫 neuroplasticity)

- **EEG**

- An amplified recording of the waves of electrical activity sweeping across the brain's surface.
- 医生在头上贴电极来观察你的**脑电波**
- 不需要记全名 考试中会给简写

- **fMRI**

- A technique for revealing blood flow and, therefore, brain activity by comparing successive MRI scans.
- 通过观察大脑**哪里在耗氧**来判断你在想什么
- 和 EEG 一样都是去测 brain activity 的，但是一个是通过 brainwave，一个是血氧

- **Lesioning**

- Tissue destruction.
- 通过**损毁**特定大脑区域来观察其功能的丧失
- 最开始的时候只能通过这些 lesioning case study 来研究大脑

1.5 - Sleep

- **Consciousness**

- Our awareness of ourselves and our environment.
- 你现在能意识到你在看这段文字，希望你 Psych 5 分~
- **Circadian rhythm**
 - The biological clock; regular bodily rhythms that occur on a 24-hour cycle.
 - 你的体温和睡眠呈现 24 小时规律波动 是一天的变化
- **Jet lag**
 - Fatigue and sleep disturbance resulting from disruption of the body's normal circadian rhythm.
 - 从北京飞往纽约后感到黑白颠倒
- **Shift work**
 - Work scheduled outside of standard daytime hours.
 - 你熬夜复习 AP 导致睡眠规律被打乱 (但这个应该叫 shift study?)
- **NREM stage 1**
 - The transition from wakefulness to sleep.
 - 刚入睡时半梦半醒的状态
- **Hypnagogic sensations**
 - Bizarre experiences, such as jerking or a feeling of falling, while transitioning to sleep.
 - 睡觉时突然感觉身体踩空蹬了一下 or 听到不存在的声音，看到闪光等
- **NREM stage 2**
 - Deeper sleep characterized by sleep spindles.
 - 睡眠中占比例最大、包含 sleep spindle 的阶段
- **NREM stage 3**
 - Deepest stage of sleep (delta waves)
 - 熟睡期，很难被叫醒，可能出现 somnambulism
- **REM sleep**
 - Rapid eye movement sleep; a recurring sleep stage during which vivid dreams commonly occur.
 - 大脑极度活跃但身体瘫痪，正在做梦
 - Active brainwaves (而且心跳呼吸会加快) + muscle paralysis
- **REM rebound**

- The tendency for REM sleep to increase following REM sleep deprivation.
- 熬夜后补觉，梦会变得特别多
- **Activation-synthesis theory**
 - The theory that dreams are the brain's attempt to synthesize random neural activity.
 - 梦是大脑在整理随机神经电信号
 - 所以这个理论在说 dreams 是没有意义的，是 random by-product, 但是 psychodynamics 觉得是 unconsciousness 的体现，所以超级有意义，要 interpretation of dreams
- **Consolidation theory**
 - Dreams help us sort out the day's events and consolidate our memories.
 - 做梦是在帮我们把记忆存进 LTM
- **Restoration of resources**
 - Sleep helps restore our bodies and immune systems.
 - 睡觉是为了恢复体力 特别是 NREM 阶段
- **Insomnia**
 - Recurring problems in falling or staying asleep.
 - 躺在床上翻来覆去睡不着
- **Narcolepsy**
 - A sleep disorder characterized by uncontrollable sleep attacks.
 - 说着说着话突然秒睡过去 可能是在情感释放之后突然发作
- **REM sleep behavior disorder**
 - A sleep disorder in which you physically act out vivid, often unpleasant dreams.
 - 在梦里打架，结果现实中也挥舞拳头踢人
- **Sleep apnea**
 - A sleep disorder characterized by temporary cessations of breathing during sleep.
 - 打呼噜时突然呼吸停了几秒
- **Somnambulism**
 - Sleepwalking.

- 在深睡期下床四处走动

1.6 - Sensation

- **Sensation**

- The process by which our sensory receptors and nervous system receive and represent stimulus energies from our environment.
- 鼻子闻到了花香

- **Transduction**

- Conversion of one form of energy into another.
- 耳朵将声波信号转化成神经电脉冲
- 是 sensation 的一部分

- **Absolute threshold**

- The minimum stimulation needed to detect a particular stimulus 50% of the time.
- 你在黑夜中能看到的微弱的一丝光
- 是 sense or not sense 的那个界限

- **Just-noticeable difference (JND)**

- The minimum difference between two stimuli required for detection 50% of the time.
- 两袋糖要相差多少克你才能感觉出一袋比另一袋重
- 基数大，需要感受到差别的区间就更大

- **Weber's law**

- To be perceived as different, two stimuli must differ by a constant minimum percentage.
- 比起总量的绝对差，我们更在意变化的百分比

- **Sensory adaptation**

- Diminished sensitivity as a consequence of constant stimulation.
- 刚进游泳池觉得冷，过一会儿就适应了水温
- 一般来说题目都在考 olfaction 的 sensory adaptation

- **Sensory interaction**

- The principle that one sense may influence another.

- 感冒鼻塞时，吃东西会觉得没味道

- **Synesthesia**

- Describing one kind of sensation in terms of another.
- 听到某种音乐时会看到特定的颜色
- 这是一种非常 rare condition 不是每个人都有

- **Retina**

- The light-sensitive inner surface of the eye.
- 眼睛里的底片，负责捕捉影像
- 里面有 rods and cones 的 photoreceptors

- **Blind spot**

- The point at which the optic nerve leaves the eye, creating a “blind” spot because no receptor cells are located there.
- 光线射到视神经出口处时，我们看不见东西
- 在这个点上，没有任何的 photoreceptor，但是大脑有 top-down processing，所以会自动补全缺失的那一点视觉信息

- **Visual nerve**

- The nerve that carries neural impulses from the eye to the brain.
- 把视觉信号传送到大脑的电缆

- **Lens**

- The transparent structure behind the pupil that changes shape to help focus images on the retina.
- 眼睛里的变焦镜头 Accommodation

- **Accommodation**

- The process by which the eye's lens changes shape to focus near or far objects on the retina.
- 看远处的山再看手中的书，晶状体在自动对焦
- 这个词在 AP Psych 有两个含义，别忘了 Piaget

- **Nearsightedness**

- Condition where far objects are blurry.

- 不带眼镜看不清黑板
- **Farsightedness**
 - Condition where near objects are blurry.
 - 老人家看报纸要拿得很远
- **Photoreceptors**
 - Cells in the retina that respond to light.
 - 感光细胞，分为视杆和视锥
- **Rods**
 - Retinal receptors that detect black, white, and gray; necessary for peripheral and twilight vision.
 - 在昏暗环境下帮你走路不撞墙
 - Peripheral vision, 所以 fovea 有很多 cones 分辨颜色，但是旁边的视线就很难分辨颜色
- **Cones**
 - Retinal receptor cells that are concentrated near the center of the retina and that function in daylight or in well-lit conditions.
 - 帮你分辨 CB 上不同 subject 的颜色 (我其实不是很喜欢 psych 的这种粉色)
 - Cones → Colors
- **Trichromatic theory**
 - The theory that the retina contains three different color receptors—one most sensitive to red, one to green, one to blue.
 - 认为我们只有 RGB 三种像素点
- **Opponent-process theory**
 - The theory that opposing retinal processes (red-green, yellow-blue, white-black) enable color vision.
 - 颜色是以对立组合的方式处理的
 - 要记得颜色的组合，会考！
- **Afterimages**
 - Images that occur when a visual sensation persists for a brief time after the original stimulus is removed.

- 盯着红色看久了，看白墙时会出现绿色的影子
- 是 opponent-process theory 的体现
- **Ganglion cells**
 - The specialized cells which lie behind the bipolar cells whose axons form the optic nerve.
 - 将初步处理后的视觉信号汇总传往大脑
- **Dichromatism**
 - A condition in which a person has only two types of cone cells.
 - 红绿色盲
- **Monochromatism**
 - A condition in which a person has only one type of cone cell and can only see in shades of gray.
 - 全色盲，世界只有黑白灰
 - 普通人是 tri，只有两种 cones 叫 di，那 mono 就是全色盲
- **Prosopagnosia**
 - Face blindness.
 - 能看清五官但认不出这人是谁
 - 它是 top-down processing 出现了问题
- **Blindsight**
 - A condition in which a person can respond to a visual stimulus without consciously experiencing it.
 - 失明者能避开地上的障碍物 neuroplasticity
- **Wavelength**
 - The distance from the peak of one light or sound wave to the peak of the next.
 - 决定了光的 hue 或声音的 pitch
 - （在这个词里找到 color 了吗
- **Pitch**
 - A tone's experienced highness or lowness; depends on frequency.
 - 女高音的高音
- **Amplitude**

- The height of a wave.
- 决定了光的 **brightness** 或声音的 **loudness**
- **Loudness**
 - A sensory characteristic of sound produced by the amplitude (intensity) of the sound wave.
 - Justin 上课时候的**超大音量**
- **Pitch perception**
 - How we hear the frequency of sound waves.
 - 大脑对**高低音**的分辨过程
- **Place theory**
 - The theory that links the pitch we hear with the place where the cochlea's membrane is stimulated.
 - 高频音刺激耳蜗的**底部**
 - 和 frequency theory 一起解释了为什么我们会听到不同的 pitch
- **Frequency theory**
 - The rate of nerve impulses traveling up the auditory nerve matches the frequency of a tone.
 - 低频音让神经纤维以**同样的频率**震动
- **Volley theory**
 - Groups of neurons fire in rapid succession to achieve higher combined frequencies.
 - 中频音通过神经元**轮班放电**来感知
- **Sound localization**
 - Determining the location of a sound source.
 - 根据声音到达左右耳的**微小时间差**判断声源
- **Conduction deafness**
 - Hearing loss caused by damage to the mechanical system that conducts sound waves to the cochlea.
 - **鼓膜破裂**导致的听力下降
 - 相对来说还是比较好介入处理的，因为神经还好着
- **Sensorineural deafness**

- Hearing loss caused by damage to the cochlea's receptor cells or to the auditory nerves.
- 长期听重金属摇滚导致毛细胞受损
- 不需要记 cochlea (耳蜗)，但需要记得是和神经相关的一种耳聋 → 人工耳蜗
- **Olfactory system**
 - The sensory system used for smelling.
 - 人类的嗅觉系统 不通过 thalamus!
- **Pheromones**
 - Chemical signals released by an animal that communicate information and affect the behavior of other animals of the same species.
 - 蚂蚁留下的气味路径 动物有的信息素
- **Gustation**
 - The sense of taste.
 - 舌头上的味觉
- **Sweet, sour, salty, bitter, umami, oleogustus**
 - The six basic tastes, which refer to the tastes of sugar, acid, salt, bitter substances, savory flavor, and fat respectively.
 - 辣不是味觉
- **Taste receptors**
 - Cells on the tongue that respond to sweet, sour, salty, bitter, umami, and oleogustus.
 - 味蕾
- **Supertasters**
 - People who have more taste buds than the average person.
 - 对味道极度敏感，通常讨厌吃苦味食物
- **Nontasters**
 - People with a low sensitivity to taste.
 - 口味重，觉得大部分菜都没味道
- **Gate control theory**

- The theory that the spinal cord contains a neurological “gate” that blocks pain signals or allows them to pass on to the brain.
- 当你撞到腿时下意识揉一揉，可以减轻痛感
- **Phantom limb**
 - Sensation that an amputated or missing limb is still attached.
 - 失去手臂的人感觉断肢处在发痒
- **Vestibular sense**
 - The sense of body movement and position, including the sense of balance.
 - 让你闭着眼也能感受到自己在旋转
 - 知道你的头和身体的相对位置，是站着还是倒立；也知道动作，比如说车在加速前进
- **Semicircular canals**
 - Three canals within the inner ear that contain specialized receptor cells that generate nerve impulses with body movement.
 - 耳蜗上方的管子，像水平仪一样维持平衡，里面有 fluid
- **Kinesthesia**
 - The system for sensing the position and movement of individual body parts.
 - 让你闭着眼也能准确摸到自己的鼻子

2.1 - Perception

- | | | |
|---|--|---|
| ☐ Bottom-up processing | ☐ Similarity | ☐ Convergence |
| ☐ Top-down processing | ☐ Attention | ☐ Monocular depth cues |
| ☐ Schema | ☐ Selective attention | ☐ Relative clarity |
| ☐ Perceptual set | ☐ Cocktail party effect | ☐ Relative size |
| ☐ Gestalt psychology | ☐ Inattentional blindness | ☐ Texture gradient |
| ☐ Closure | ☐ Change blindness | ☐ Linear perspective |
| ☐ Figure and ground | ☐ Binocular depth cues | ☐ Interposition |
| ☐ Proximity | ☐ Retinal disparity | ☐ Apparent movement |

2.2 - Thinking, Problem-Solving, Judgments, and Decision Making

- | | | |
|---|---|---|
| ☐ Prototypes | ☐ Availability heuristic | ☐ Executive functions |
| ☐ Assimilation | ☐ Mental set | ☐ Creativity |
| ☐ Accommodation | ☐ Priming | ☐ Divergent thinking |
| ☐ Algorithms | ☐ Framing | ☐ Convergent thinking |
| ☐ Heuristics | ☐ Gambler's fallacy | ☐ Functional fixedness |
| ☐ Representativeness heuristic | ☐ Sunk-cost fallacy | |

2.3 - Introduction to Memory

- | | | |
|---|---|---|
| ☐ Explicit memory | ☐ Primary memory system | ☐ Echoic memory |
| ☐ Episodic memory | ☐ Central executive | ☐ Automatic processing |
| ☐ Semantic memory | ☐ Phonological loop | ☐ Effortful processing |
| ☐ Implicit memory | ☐ Visuospatial sketchpad | ☐ Storage |
| ☐ Procedural memory | ☐ Long-term memory | ☐ Levels of processing model |
| ☐ Prospective memory | ☐ Multi-store model | ☐ Shallow encoding |
| ☐ Long-term potentiation | ☐ Sensory memory | ☐ Deep encoding |
| ☐ Working memory model | ☐ Iconic memory | ☐ Structural, phonemic, semantic |

2.4 - Encoding Memories

- | | | |
|---|--|---|
| ☐ Encoding | ☐ Hierarchies (as they pertain to grouping) | ☐ Distributed practice |
| ☐ Mnemonic devices | ☐ Spacing effect | ☐ Serial position effect |
| ☐ Method of loci | ☐ Memory consolidation | ☐ Primacy effect |
| ☐ Chunking | ☐ Massed practice | ☐ Recency effect |
| ☐ Categories (as they pertain to grouping) | | |

2.5 - Storing Memories

- | | | |
|--|--|--|
| ☐ Sensory memory | ☐ Maintenance rehearsal | ☐ Autobiographical memory |
| ☐ Short-term memory | ☐ Elaborative rehearsal | ☐ Anterograde amnesia |
| ☐ Working memory | ☐ Memory retention | ☐ Alzheimer's disease |
| ☐ Long-term memory | ☐ Retrograde amnesia | ☐ Infantile amnesia |

2.6 - Retrieving Memories

- | | | |
|--------------------------------------|---|---|
| ☐ Retrieval | ☐ Retrieval cues | ☐ State-dependent memory |
| ☐ Recall | ☐ Context-dependent memory | ☐ Testing effect |
| ☐ Recognition | ☐ Mood-congruent memory | ☐ Metacognition |

2.7 - Forgetting and Other Memory Challenges

- | | | |
|---|---|--|
| ☐ Forgetting curve | ☐ Tip-of-the-tongue phenomenon | ☐ Source amnesia |
| ☐ Encoding failure | ☐ Repression (psychodynamic) | ☐ Constructive memory |
| ☐ Proactive interference | ☐ Misinformation effect | ☐ Imagination inflation |
| ☐ Retroactive interference | | |

2.8 - Intelligence and Achievement

- | | | |
|---|--|--|
| ☐ Intelligence | ☐ Construct validity | ☐ Stereotype lift |
| ☐ g (general intelligence) | ☐ Predictive validity | ☐ Flynn Effect |
| ☐ Intelligence quotient (IQ) | ☐ Reliability | ☐ Achievement tests |
| ☐ Mental age | ☐ Test-retest reliability | ☐ Aptitude tests |
| ☐ Chronological age | ☐ Split-half reliability | ☐ Fixed mindset |
| ☐ Standardization | ☐ Stereotype threat | ☐ Growth mindset |
| ☐ Validity | | |

2.1 - Perception

- **Bottom-up processing**
 - Processing that begins with raw sensory data and builds up to perception — **no prior knowledge** involved.
 - 老外第一次看汉字，一笔一画地拆解，因为没有任何已有知识可以依赖
- **Top-down processing**
 - Using **prior knowledge, expectations, and context** to interpret sensory information.
 - 研究表明汉字的序顺并不一定影响阅读，你的大脑自动用已有知识补全了意思
- **Schema**
 - A mental framework that organizes knowledge and guides expectations about the world.
 - 你对餐厅的schema：进门、点餐、吃饭、买单，去到任何餐厅都会这样预期
- **Perceptual set**
 - A mental predisposition to perceive things in a certain way, influenced by experience, context, culture, emotions, and motivation. **Not inherently positive or negative** — differs from mental set.
 - 先入为主，你刚看完恐怖片，走在黑暗的路上，树影都觉得像鬼
- **Gestalt psychology**
 - A school of psychology emphasizing that **the whole is greater than the sum of its parts** — the brain organizes sensory information into meaningful wholes.
 - 我们看到的不是一堆孤立的点和线，而是一个完整的图形
- **Closure**
 - The tendency to mentally **fill in gaps** in incomplete figures to perceive them as complete.
 - 一个缺了一块的圆，你的大脑会自动脑补成完整的圆
- **Figure and ground**
 - The brain automatically separates visual input into a main object (**figure**) and its background (**ground**).
 - 经典的花瓶与人脸图，你要么看到花瓶，要么看到两张对视的脸
- **Proximity**
 - Objects that are **close together** in space tend to be perceived as belonging to the same group.
 - 三个点靠在一起、另外三个点靠在一起，你会自动把它们看成两组
- **Similarity**
 - Objects that **look alike** (color, shape, size) tend to be grouped together.
 - 一排红点和一排蓝点交替排列，你会按颜色把它们分成不同的组

- **Attention**
 - The process of focusing conscious awareness on a particular stimulus while filtering out others.
 - 你在教室里专心听老师讲课，忽略了窗外的鸟叫声
- **Selective attention**
 - Focusing on one specific stimulus while ignoring others — acts as a **filter**.
 - 你在嘈杂的自习室里专心做题，周围的聊天声你完全没在意
- **Cocktail party effect**
 - The ability to detect personally relevant information (like your name) even when not actively attending to it — suggests background monitoring.
 - 嘈杂的派对上你不在听别人聊天，但远处有人叫你名字，你立刻注意到了
- **Inattention blindness**
 - Failing to notice a fully visible but unexpected stimulus when attention is focused elsewhere.
 - 经典大猩猩实验：你专注数传球次数，完全没看到一只大猩猩走过画面
- **Change blindness**
 - Failing to detect obvious changes in a scene when the change occurs during a brief disruption.
 - 问路实验：你正在给一个人指路，中途被换成另一个人，你竟然没发现
- **Binocular depth cues**
 - Depth cues that require **both eyes** working together, mainly for judging nearby distances.
 - 需要两只眼睛才能获得的深度信息
- **Retinal disparity**
 - The slight difference in images projected on the left and right retinas; **greater disparity = closer object**.
 - 用左眼和右眼分别看手指，位置不一样，手指越近差异越大
- **Convergence**
 - The inward turning of eyes when viewing a nearby object; **more muscle tension = closer object**.
 - 看很近的东西时眼睛会变成斗鸡眼，大脑通过眼部肌肉紧张程度判断远近
- **Monocular depth cues**
 - Depth cues available to **each eye alone**, useful for perceiving distance in 2D images.
 - 用一只眼睛看照片也能判断远近的线索
- **Relative clarity**
 - Clearer objects appear **closer**; hazier objects appear **farther** away.
 - 远处的山看起来雾蒙蒙的，近处的树看起来很清晰

- **Relative size**
 - When two objects are assumed to be the same actual size, the **smaller image** appears farther away.
 - 两辆一样的车，看起来小的那辆离你更远
- **Texture gradient**
 - Nearby surfaces have **detailed, coarse** textures; distant surfaces appear **dense and smooth**.
 - 脚下的草地纹理清晰可见，远处的草地纹理变得密集模糊
- **Linear perspective**
 - Parallel lines appear to **converge** in the distance; more convergence = farther.
 - 铁轨在远方汇聚成一个点
- **Interposition**
 - When one object **overlaps/blocks** another, the blocked object is perceived as farther away.
 - 一座楼挡住了另一座楼，被挡住的那座看起来更远
- **Apparent movement**
 - The perception of movement from a rapid series of **still images** — the basis of film and animation.
 - 电影本质就是一系列静止图片快速播放，让你觉得画面在动

2.2 - Thinking, Problem-Solving, Judgments, and Decision Making

- **Prototypes**
 - The **most typical** example of a concept, used as a mental reference for categorization.
 - 提到鸟，你脑中浮现的是麻雀而不是企鹅，因为麻雀更符合鸟的典型特征
- **Assimilation**
 - Fitting new information **into an existing schema** without changing the schema.
 - 小孩知道狗是四条腿会叫的动物，看到新品种的狗也直接归类为狗
- **Accommodation**
 - **Modifying an existing schema** to incorporate new information that doesn't fit.
 - 小孩把猫也叫狗，后来发现猫会喵喵叫，于是调整schema区分猫和狗
- **Algorithms**
 - A step-by-step procedure that **guarantees** a correct solution, but may be slow.
 - 超市找薯片，一排一排货架挨个找，肯定能找到但很慢
- **Heuristics**
 - Mental shortcuts that allow quick judgments; efficient but **not guaranteed** to be correct.
 - 超市找薯片，直接去零食区找，快但可能走错区域
- **Representativeness heuristic**
 - Judging the likelihood of something based on how well it **matches a prototype**, often ignoring base rates.
 - 一个安静爱读书的人，你觉得他是图书管理员而不是农民，但其实农民人数远多于图书管理员
- **Availability heuristic**
 - Judging frequency or probability based on how **easily examples come to mind**.
 - 新闻总报飞机失事，所以你觉得坐飞机比坐车危险，但其实车祸死亡率更高
- **Mental set**
 - The tendency to use **previously successful strategies** even when they are no longer effective — leads to rigid thinking.
 - 你一直用某个数学公式解题，遇到新类型的题还是下意识套用，结果错了
- **Priming**
 - Exposure to a prior stimulus **unconsciously influences** the response to a subsequent stimulus.
 - 先看到香蕉这个词，然后填 Y_L__，你更可能写出 YELLOW
- **Framing**
 - The way information is **presented** significantly influences decisions and judgments.
 - 手术存活率90%听起来很安全，死亡率10%听起来很危险，其实是同一件事

- **Gambler's fallacy**
 - The false belief that past random events affect future probabilities of **independent** events.
 - 连续抛了9次正面，你觉得下一次一定是反面，但每次概率都是50%
- **Sunk-cost fallacy**
 - Continuing an endeavor because of **previously invested resources**, even when it's no longer rational.
 - 电影很难看，但已经花了钱买票，就强迫自己看完
- **Executive functions**
 - Higher-order cognitive processes (in the **frontal lobes**) that manage attention, planning, decision-making, and self-regulation.
 - 你在写作业时控制自己不去刷手机，这就是执行功能在工作
- **Creativity**
 - The ability to produce ideas that are both **novel and valuable**.
 - 用一块砖头想出几十种用途：盖房子、当哑铃、当画板.....
- **Divergent thinking**
 - Thinking that **explores multiple possible solutions** — open-ended and creative.
 - 头脑风暴：一块砖头有多少种用途？天马行空地想
- **Convergent thinking**
 - Thinking that **narrows down to one correct answer** — logical and focused.
 - AP选择题四选一，你需要排除错误选项，找出唯一正确答案
- **Functional fixedness**
 - The inability to see an object as having a function **beyond its typical use** — a type of mental set.
 - 蜡烛问题：你想不到把图钉盒倒空当烛台，因为你被它装图钉的功能固着了

2.3 - Introduction to Memory

- **Explicit memory**

- Memories that can be **consciously recalled** and verbally declared. Also called **declarative memory**.
- 你能清楚说出昨天晚餐吃了什么，中国的首都是北京

- **Episodic memory**

- A type of explicit memory for **personal experiences** tied to specific times and places — like life episodes.
- 你记得去年生日派对的场景、声音和当时的感受

- **Semantic memory**

- A type of explicit memory for **general facts and knowledge**, independent of personal experience.
- 地球是圆的， $1+1=2$ ，这些知识不依赖于你的个人经历

- **Implicit memory**

- Memories that influence behavior **without conscious awareness**.
- 你骑自行车时不需要有意识地想每一步该怎么做

- **Procedural memory**

- A type of implicit memory for **motor skills and habits** — "how-to" knowledge.
- 骑车、打字、弹钢琴，你会做但说不清楚具体怎么协调的

- **Prospective memory**

- Remembering to **perform a planned action in the future** — "remembering to remember."
- 记住下午三点要去取快递

- **Long-term potentiation**

- The **strengthening of synaptic connections** through repeated stimulation — the biological basis of learning and memory.
- 这世上本没有路，走的人多了也便成了路；反复复习就是在强化神经通路

- **Working memory model**

- An updated model of STM emphasizing **active processing**, consisting of the central executive, phonological loop, and visuospatial sketchpad.
- 短期记忆的升级版，不只是被动存放信息，而是主动加工

- **Primary memory system**

- An early term for short-term/working memory — the system that **actively holds** information currently in use.
- 你正在使用的临时工作台，比如心算时暂时记住中间结果

- **Central executive**
 - The component of working memory that **directs attention** and coordinates the phonological loop and visuospatial sketchpad.
 - 老板角色，分配注意力，协调其他两个子系统的工作
- **Phonological loop**
 - The component of working memory that processes **auditory and verbal** information — the "inner ear and voice."
 - 你在脑海里默念一个电话号码来记住它
- **Visuospatial sketchpad**
 - The component of working memory that processes **visual and spatial** information — the "inner eye."
 - 你在脑中想象一张地图或者旋转一个三维物体
- **Long-term memory**
 - The relatively **permanent and unlimited** storage system for knowledge, skills, and experiences.
 - 你多年前学会的母语、骑车技能、童年回忆都存在这里
- **Multi-store model**
 - Atkinson-Shiffrin model proposing three sequential stages: **sensory memory → short-term memory → long-term memory**.
 - 信息先进入感觉记忆，被注意后进入短期记忆，经过复述后存入长期记忆
- **Sensory memory**
 - A brief storage (**≤1 second**) that holds large amounts of raw sensory input before it fades or is attended to.
 - 你快速晃动烟花棒时看到的完整光轨，就是视觉感觉记忆
- **Iconic memory**
 - **Visual** sensory memory, lasting less than a second.
 - 晃动烟花棒看到完整光轨，闭眼后画面短暂停留
- **Echoic memory**
 - **Auditory** sensory memory, lasting about 3-4 seconds.
 - 老师说了一句话你没听清，但几秒内可以在脑中回放刚才的声音
- **Automatic processing**
 - Encoding of information **without conscious effort** — such as space, time, and frequency.
 - 你不用刻意去记，就知道今天在路上碰到了好几次同一个朋友
- **Effortful processing**
 - Encoding that requires **conscious attention and deliberate effort**.
 - 背单词、复习公式，需要你集中注意力反复练习

- **Storage**
 - The process of **maintaining** encoded information in memory over time.
 - 把文件保存到硬盘上
- **Levels of processing model**
 - Craik & Lockhart's theory that memory depends on the **depth of processing** — deeper processing leads to better memory.
 - 加工越深记得越牢，理解意义比死记字形效果好得多
- **Shallow encoding**
 - Processing information at a **surface level** (e.g., appearance or sound), producing weak memory.
 - 只注意到 Peter Parker 是大写的，或者有几个字母
- **Deep encoding**
 - Processing information at a **meaningful level** (semantics), producing strong and lasting memory.
 - 看到 Peter Parker 想到他是蜘蛛侠，和已有知识建立联系
- **Structural, phonemic, semantic**
 - Three levels: **structural** = physical features (shallowest), **phonemic** = sound, **semantic** = meaning (**deepest, most effective**).
 - Peter Parker: 结构层看字母形状，语音层注意到两个P押头韵，语义层想到他是蜘蛛侠

2.4 - Encoding Memories

- **Encoding**
 - The process of **converting** sensory information into a form that can be stored in memory.
 - 用键盘打字，把外界信息变成电子信号存进电脑
- **Mnemonic devices**
 - Memory aids that use **strategies** (imagery, organization, acronyms) to enhance encoding and recall.
 - 用 My Very Educated Mother Just Served Us Noodles 记太阳系行星顺序
- **Method of loci**
 - A mnemonic technique that links items to **specific locations** in a familiar physical space — "memory palace."
 - 把要记的东西放在你家里不同位置，回忆时在脑中走一遍就能看到它们
- **Chunking**
 - Grouping individual pieces of information into **larger, meaningful units** to overcome STM capacity limits (7 ± 2).
 - 手机号 13812345678 分成 138-1234-5678 三组来记
- **Categories (as they pertain to grouping)**
 - Organizing information into **meaningful groups** based on shared characteristics to aid encoding.
 - 购物清单分成乳制品、蔬菜、零食等类别来记
- **Hierarchies (as they pertain to grouping)**
 - Organizing information in a **structured, ranked order** from broad to specific to enhance memory.
 - 生物分类：动物→哺乳动物→犬科→狗，层层细分更好记
- **Spacing effect**
 - **Distributed practice** over time produces better long-term retention than massed practice.
 - 每天复习30分钟，连续一周，比考前通宵抱佛脚效果好得多
- **Memory consolidation**
 - The process of converting fragile short-term memories into **stable long-term memories**, involving the **hippocampus**.
 - 电脑把RAM里的临时数据写入硬盘永久保存，睡眠对巩固记忆很重要
- **Massed practice**
 - Cramming — studying intensively in **one concentrated session**; poor for long-term retention.
 - 考试前一晚通宵复习，短期可能有用但很快就忘

- **Distributed practice**
 - Spreading study sessions **over time**; leads to stronger and more durable memory.
 - 每天学一点，连续一周复习，记忆效果远好于临时抱佛脚
- **Serial position effect**
 - The tendency to best recall items at the **beginning and end** of a list, with poorest recall in the middle.
 - 背一长串单词，最容易记住开头和结尾的，中间的最容易忘
- **Primacy effect**
 - Better recall for items at the **beginning** of a list — due to more rehearsal time, transferring them to LTM.
 - 列表开头的词你有更多时间反复默念，所以记得牢
- **Recency effect**
 - Better recall for items at the **end** of a list — because they are still in working memory.
 - 列表最后几个词刚刚才听到，还留在短期记忆里所以容易想起

2.5 - Storing Memories

- **Sensory memory**

- The initial, brief storage of sensory information; **large capacity but very short duration** ($\leq 1\text{s}$ for iconic, $\sim 3\text{-}4\text{s}$ for echoic).
- 记忆的接待大厅，所有感官信息先到这里，绝大部分迅速消失

- **Short-term memory**

- Temporary storage with **limited capacity (7 ± 2 items)** and **limited duration ($\sim 20\text{-}30$ seconds)** without rehearsal.
- 临时记一个验证码，不复述的话30秒内就忘了

- **Working memory**

- The active processing system (upgraded STM) that **manipulates and processes** information, not just holds it.
- 你做心算时，一边记住中间结果一边计算下一步

- **Long-term memory**

- The relatively **permanent** storage with virtually **unlimited capacity and duration**.
- 你的知识、技能和所有经历的永久储存库

- **Maintenance rehearsal**

- **Simple repetition** to keep information in STM; does not effectively transfer to LTM.
- 不停念叨验证码 583926，但挂了电话马上就忘

- **Elaborative rehearsal**

- Connecting new information to **existing knowledge** through meaningful, deep processing — effective for LTM transfer.
- 购物清单按乳制品、清洁用品分类来记，把新信息和已有知识深度连接

- **Memory retention**

- The ability to **maintain** stored information over time.
- 学过的知识过了一个月还能记得多少

- **Retrograde amnesia**

- Inability to recall memories from **before** a traumatic event. **Retro = backward**.
- 车祸后醒来，不记得自己是谁，不认识家人，丢失的是过去的记忆

- **Autobiographical memory**

- Personal memory about one's own life, combining both **episodic and semantic** memories. Key: **better memory about yourself**.
- 关于你自己人生故事的记忆，包括个人经历和关于自己的事实

- **Anterograde amnesia**

- Inability to form **new** long-term memories after a traumatic event. **Antero = forward**.
- Henry Molaison切除海马体后，旧记忆都在，但你离开房间再回来他就不记得见过你

- **Alzheimer's disease**

- A progressive **neurodegenerative** disease characterized by memory loss, cognitive decline, and behavioral changes.
- 从忘记最近的事开始，逐渐不认识家人、忘记个人信息、丧失自理能力

- **Infantile amnesia**

- The inability to recall personal experiences from **before age 3-4**, due to immature hippocampus, incomplete language, and undeveloped sense of self.
- 你几乎想不起3岁前发生过什么事

2.6 - Retrieving Memories

- **Retrieval**
 - The process of **accessing and bringing stored information** back into conscious awareness.
 - 从记忆库中找到并调出你需要的信息
- **Recall**
 - Retrieving information **without** external cues — more difficult.
 - 填空题和简答题，没有提示，你得自己从记忆中把答案找出来
- **Recognition**
 - Identifying previously learned information when it is **presented again** — easier than recall.
 - 选择题，正确答案就在眼前，你只需要认出它
- **Retrieval cues**
 - Stimuli that help **trigger** the recall of stored memories; can be external (scene, smell) or internal (mood, thought).
 - 闻到某种香水的味道，突然想起一个老朋友
- **Context-dependent memory**
 - Memory retrieval is better when the **physical environment** at encoding and retrieval match.
 - 在同一间教室学习和考试，你可能更容易想起复习过的内容
- **Mood-congruent memory**
 - Current mood influences **which memories** are recalled — we recall memories consistent with our present mood.
 - 心情好的时候容易想起开心的事，悲伤的时候满脑子都是难过的回忆
- **State-dependent memory**
 - Memory retrieval is better when the **physiological/psychological state** at encoding and retrieval match.
 - 学习时喝了咖啡处于兴奋状态，考试前也喝一杯可能帮你更好地回忆
- **Testing effect**
 - **Self-testing** (retrieval practice) strengthens long-term memory more effectively than simply re-reading material.
 - 做练习题、考自己、向别人解释概念，比反复看书效果好得多
- **Metacognition**
 - **Thinking about your own thinking** — monitoring and reflecting on your cognitive processes.
 - 你意识到自己没听懂老师讲的内容，决定换一种方式学习

2.7 - Forgetting and Other Memory Challenges

- **Forgetting curve**

- Ebbinghaus's finding that forgetting occurs rapidly after learning (~70% lost within 24 hours) and then levels off. Countered by **distributed practice**.
- 新知识学完24小时内忘掉大部分，不复习的话考完试就还给老师了

- **Encoding failure**

- Information was **never properly encoded** into LTM in the first place — not truly forgetting, but never stored.
- 你天天用硬币但说不出硬币上的头像朝哪边，因为从来没认真编码过

- **Proactive interference**

- **Old** memories interfere with the retrieval of **new** information. **Pro = forward** (old → new).
- 搬家了填表格时下意识写出旧地址，旧记忆干扰了新记忆

- **Retroactive interference**

- **New** memories interfere with the retrieval of **old** information. **Retro = backward** (new → old).
- 学了一首新曲子后，发现想不起以前那首旧曲子的旋律了

- **Tip-of-the-tongue phenomenon**

- The feeling of **knowing** information but being temporarily unable to retrieve it — available but not retrievable.
- 那个人叫什么来着.....就在嘴边就是说不出来，过一会突然想起来了

- **Repression (psychodynamic)**

- The **unconscious** pushing of painful or traumatic memories out of awareness as a defense mechanism. **Controversial** — lacks strong scientific support.
- 弗洛伊德认为人会无意识地将创伤记忆排除出意识来保护自己，但这个概念在现代心理学中有争议

- **Misinformation effect**

- Exposure to **misleading post-event information** distorts the original memory.
- 车祸实验：用 smashed 代替 hit 来提问，参与者估计的车速更高，还声称看到了不存在的碎玻璃

- **Source amnesia**

- Remembering information but **forgetting where, when, or how** it was learned.
- 你记得一个冷知识但想不起是从哪知道的，既视感（déjà vu）可能也与此有关

- **Constructive memory**

- Memory is not simply retrieved but **actively reconstructed** each time, filling gaps with existing knowledge, beliefs, and expectations.
- 每次回忆都是一次重新建构，你会用已有经验自动填补记忆中的空白

- **Imagination inflation**

- **Imagining** a fictitious event increases confidence that it actually happened.
- 反复想象自己小时候在商场走丢，即使从未发生，你可能会觉得这事真的发生过

2.8 - Intelligence and Achievement

- **Intelligence**
 - The ability to **learn from experience, solve problems, and adapt** to new situations; considered adaptive, involving multiple mental processes, and culture-specific.
 - 没有统一定义，但一般认为是学习能力、解决问题和适应新环境的能力
- **g (general intelligence)**
 - Spearman's concept of a single **underlying general intelligence factor** that influences performance across all cognitive tasks.
 - 好学生啥都好，一项认知测试强的人其他方面往往也不错
- **Intelligence quotient (IQ)**
 - A standardized score representing intellectual ability, with **100 as the average**, following a normal distribution.
 - 智商分数，100分为平均值，用来和同龄人比较
- **Mental age**
 - The age level of intellectual performance a person demonstrates on a test, **regardless of actual age**.
 - 一个8岁孩子的测试表现和平均10岁孩子一样好，他的心理年龄就是10岁
- **Chronological age**
 - A person's **actual age** in years from birth.
 - 你的实际年龄，用来和心理年龄对比计算IQ
- **Standardization**
 - Establishing **uniform procedures** for administering and scoring a test, with norms from a representative sample.
 - 所有考生在相同条件下考试，用统一的评分标准改卷，分数才有可比性
- **Validity**
 - The extent to which a test **measures what it claims to measure**.
 - 不能站在体重秤上测身高，体重秤测身高就是无效的
- **Construct validity**
 - Whether a test accurately measures the **theoretical psychological concept** it intends to measure (e.g., anxiety, intelligence).
 - 一个智力测试是否真的在测量智力这个心理学概念
- **Predictive validity**
 - Whether test scores can effectively **predict future behavior** or performance.
 - SAT分数能否预测大学GPA

- **Reliability**
 - The extent to which a test yields **consistent, reproducible** results.
 - 能力没变化的情况下，测两次应该得到相近的分数
- **Test-retest reliability**
 - Consistency of scores when the **same test is given twice** to the same group at different times.
 - 同一组人隔一个月测两次，看同一个人的分数是否接近
- **Split-half reliability**
 - Consistency between **two halves** of a single test (e.g., odd vs. even items).
 - 把试卷的奇数题和偶数题分数做比较，高相关说明在测量同样的东西
- **Stereotype threat**
 - Anxiety that one's performance will **confirm a negative stereotype** about their group, which impairs performance.
 - 女生数学不好的刻板印象让女生在数学考试时更紧张，反而考得更差
- **Stereotype lift**
 - **Improved** performance resulting from awareness of a positive stereotype about one's group.
 - 亚洲人数学好的刻板印象让亚洲学生在数学考试时更自信
- **Flynn Effect**
 - The observed trend that average IQ scores have been **rising across generations** worldwide — requiring periodic restandardization.
 - 全球IQ平均分一代比一代高，可能因为营养、教育和科技进步
- **Achievement tests**
 - Tests that measure **what a person has already learned**.
 - AP考试、期末考试、单元测验
- **Aptitude tests**
 - Tests designed to **predict future learning ability** or potential, not current knowledge.
 - SAT预测你未来在大学的学习潜力
- **Fixed mindset**
 - The belief that intelligence is **innate and unchangeable**; failure = lack of ability.
 - 觉得智力天生固定，害怕挑战，因为失败意味着自己不聪明
- **Growth mindset**
 - The belief that intelligence can be **developed through effort and learning**; failure = opportunity to grow.
 - 觉得能力可以通过努力提升，失败是学习和成长的机会

3.1 - Themes and Methods in Developmental Psychology

- | | | |
|--|--|---|
| ☐ Chronological development | ☐ Discontinuous development | ☐ Stability and change |
| ☐ Continuous development | ☐ Lifespan development | ☐ Nature and nurture |

3.2 - Physical Development Across the Lifespan

- | | | |
|---|--|--|
| ☐ Teratogens | ☐ Visual cliff | ☐ Primary sex characteristics |
| ☐ Fine motor coordination | ☐ Critical periods | ☐ Secondary sex characteristics |
| ☐ Gross motor coordination | ☐ Sensitive periods | ☐ Menarche |
| ☐ Maturation | ☐ Imprinting | ☐ Spermatarche |
| ☐ Reflexes | ☐ Growth spurt | ☐ Menopause |
| ☐ Rooting reflex | ☐ Puberty | |

3.3 - Gender and Sexual Orientation

- ☐ (No specific terms for this topic)

3.4 - Cognitive Development Across the Lifespan

- | | | |
|---|--|---|
| ☐ Schemas | ☐ Mental symbols | ☐ Abstract thinking |
| ☐ Assimilation | ☐ Pretend play | ☐ Hypothetical thinking |
| ☐ Accommodation | ☐ Conservation | ☐ Scaffolding (as it pertains to Vygotsky) |
| ☐ Sensorimotor stage | ☐ Reversibility | ☐ Zone of proximal development |
| ☐ Preoperational stage | ☐ Animism | ☐ Crystallized intelligence |
| ☐ Concrete operational stage | ☐ Egocentrism | ☐ Fluid intelligence |
| ☐ Formal operational stage | ☐ Theory of mind | ☐ Dementia |
| ☐ Object permanence | ☐ Systematic thinking | |

3.5 - Communication and Language Development

- | | | |
|------------------------------------|-----------------------------------|---|
| ☐ Phonemes | ☐ Syntax | ☐ One-word stage |
| ☐ Morphemes | ☐ Cooin | ☐ Telegraphic speech |
| ☐ Semantics | ☐ Babbling | ☐ Overgeneralization of language rules |
| ☐ Grammar | | |

3.6 - Social-Emotional Development Across the Lifespan

- | | | |
|--|---|--|
| ☐ Ecological systems theory | ☐ Disorganized attachment | ☐ Intimacy and isolation |
| ☐ Microsystem | ☐ Separation anxiety | ☐ Generativity and stagnation |
| ☐ Mesosystem | ☐ Temperament | ☐ Integrity and despair |
| ☐ Exosystem | ☐ Parallel play | ☐ Adverse childhood experiences (ACEs) |
| ☐ Macrosystem | ☐ Imaginary audience | ☐ Achievement (adolescent identity development) |
| ☐ Chronosystem | ☐ Personal fable | ☐ Diffusion (adolescent identity development) |
| ☐ Authoritarian parenting | ☐ Social clock | ☐ Foreclosure (adolescent identity development) |
| ☐ Authoritative parenting | ☐ Emerging adulthood | ☐ Moratorium (adolescent identity development) |
| ☐ Permissive parenting | ☐ Stage theory of psychosocial development (Erikson) | ☐ Racial/ethnic identity |
| ☐ Attachment styles | ☐ Trust and mistrust | ☐ Sexual orientation |
| ☐ Secure attachment | ☐ Autonomy and shame and doubt | ☐ Religious identity |
| ☐ Insecure attachment | ☐ Initiative and guilt | ☐ Occupational identity |
| ☐ Avoidant attachment | ☐ Industry and inferiority | ☐ Familial identity |
| ☐ Anxious attachment | ☐ Identity and role confusion | ☐ Possible selves |

3.7 - Classical Conditioning

- | | | |
|---|--|--|
| ☐ Behavioral perspective | ☐ Conditioned stimulus | ☐ Counterconditioning |
| ☐ Classical conditioning | ☐ Conditioned response | ☐ Taste aversion |
| ☐ Association | ☐ Extinction | ☐ One-trial conditioning |
| ☐ Acquisition | ☐ Spontaneous recovery | ☐ One-trial learning |
| ☐ Associative learning | ☐ Stimulus discrimination | ☐ Biological preparedness |
| ☐ Unconditioned stimulus | ☐ Stimulus generalization | ☐ Habituation |
| ☐ Unconditioned response | ☐ Higher-order conditioning | |

3.8 - Operant Conditioning

- | | | |
|---|---|---|
| ☐ Operant conditioning | ☐ Secondary reinforcer | ☐ Continuous reinforcement |
| ☐ Reinforcement | ☐ Reinforcement discrimination | ☐ Partial reinforcement |
| ☐ Punishment | ☐ Reinforcement generalization | ☐ Fixed interval |
| ☐ Law of Effect | ☐ Shaping | ☐ Variable interval |
| ☐ Superstitious behavior | ☐ Instinctive drift | ☐ Fixed ratio |
| ☐ Learned helplessness | ☐ Negative reinforcement | ☐ Variable ratio |
| ☐ Positive reinforcement | ☐ Reinforcement schedules | ☐ Scalloped graph |
| ☐ Primary reinforcers | | |

3.9 - Social, Cognitive, and Neurological Factors in Learning

- | | | |
|---|---|--|
| ☐ Social learning theory | ☐ Modeling | ☐ Latent learning |
| ☐ Vicarious conditioning | ☐ Insight learning | ☐ Cognitive maps |

3.1 - Themes and Methods in Developmental Psychology

- **Chronological development**
 - Studying development by following **age-related changes** over time.
 - 按年龄顺序研究人从婴儿到老年的变化
- **Continuous development**
 - Development is a **gradual, smooth** process without distinct stages.
 - 像一个斜坡，能力慢慢增长，没有突然的跳跃
- **Discontinuous development**
 - Development occurs in **distinct stages**, each qualitatively different from the last.
 - 像爬楼梯，每上一层就有质的变化
- **Lifespan development**
 - Development is a process that continues **from conception to death**, not just childhood.
 - 人一辈子都在发展，不只是小时候
- **Stability and change**
 - The debate over whether traits remain consistent or change across the lifespan.
 - 小时候内向的人长大了还会内向吗？
- **Nature and nurture**
 - The debate over the relative contributions of **genetics** vs. **environment** to development.
 - 智商是天生的还是后天培养的？两者都有影响

3.2 - Physical Development Across the Lifespan

- **Teratogens**
 - Harmful agents (drugs, alcohol, viruses) that can cause birth defects during **prenatal** development.
 - 孕妇喝酒可能导致胎儿酒精综合征
- **Fine motor coordination**
 - Control of **small muscles**, especially in the hands and fingers.
 - 用筷子、写字、系鞋带
- **Gross motor coordination**
 - Control of **large muscles** used for walking, running, and jumping.
 - 走路、跑步、跳跃
- **Maturation**
 - The **biologically driven** sequence of growth and development that is relatively unaffected by experience.
 - 所有婴儿都是先会坐再会走，这个顺序不需要教
- **Reflexes**
 - Involuntary, **automatic responses** to stimuli that are present at birth.
 - 新生儿天生就有的自动反应，比如抓握反射
- **Rooting reflex**
 - When a newborn's cheek is touched, it turns toward the stimulus and opens its mouth to feed.
 - 碰婴儿脸颊，他会自动把头转过来找奶吃
- **Visual cliff**
 - A laboratory device used to test **depth perception** in infants; most infants avoid the "deep" side by 6-14 months.
 - 把婴儿放在玻璃桌上，下面看起来是悬崖，婴儿不敢爬过去，说明已有深度知觉
- **Critical periods**
 - A **strict** time window during which certain experiences **must** occur for normal development; missing it causes irreversible effects.
 - 如果孩子在7岁前没有接触语言，语言能力可能永久受损
- **Sensitive periods**
 - An **optimal** but not absolute time window when the organism is especially responsive to certain experiences.
 - 幼儿期学语言最容易，但成年后也不是完全学不会

- **Imprinting**
 - A rapid form of attachment in certain animals during a critical period, typically to the **first moving object** they see; **not found in humans**.
 - 小鹅出生后把第一个看到的移动物体当妈，这是动物特有的，人类没有印刻
- **Growth spurt**
 - A period of **rapid physical growth**, especially during puberty.
 - 青春期突然长高了很多
- **Puberty**
 - The period of physical maturation involving hormonal surges and the development of **sexual maturity**.
 - 青春期身体开始发育，出现各种生理变化
- **Primary sex characteristics**
 - Body structures directly involved in **reproduction** (e.g., ovaries, testes, uterus).
 - 和生殖直接相关的器官，如卵巢、子宫、睾丸
- **Secondary sex characteristics**
 - Physical features **not directly** involved in reproduction that appear at puberty.
 - 男生长喉结和胡须，女生乳房发育
- **Menarche**
 - A female's **first menstrual period**, signaling reproductive maturity.
 - 女生第一次来月经
- **Spermarche**
 - A male's **first ejaculation**, signaling reproductive maturity.
 - 男生第一次遗精
- **Menopause**
 - The cessation of menstruation in women, typically around age **50**, ending reproductive capacity.
 - 女性大约50岁左右停经，伴随荷尔蒙变化和各种身心不适

3.4 - Cognitive Development Across the Lifespan

- **Schemas**
 - Mental frameworks or concepts that help us **organize and interpret** information.
 - 小孩脑中有一个「狗」的框架：四条腿、毛茸茸、会叫
- **Assimilation**
 - Interpreting new experiences in terms of **existing** schemas.
 - 小孩第一次看到猫，叫它「狗狗」，因为也是四条腿毛茸茸的
- **Accommodation**
 - **Modifying** existing schemas or creating new ones to incorporate new information.
 - 小孩发现猫会喵喵叫，和狗不一样，于是建立了新的「猫」的概念
- **Sensorimotor stage**
 - Piaget's **Stage 1** (birth to ~2 years): infants learn through senses and motor actions; develop **object permanence**.
 - 婴儿通过抓、咬、看来认识世界，活在当下
- **Preoperational stage**
 - Piaget's **Stage 2** (~2-7 years): symbolic thinking develops but children lack **conservation** and are **egocentric**.
 - 会用语言和假装游戏，但认为把水倒进高杯子水就变多了
- **Concrete operational stage**
 - Piaget's **Stage 3** (~7-11 years): children gain **logical thinking** about concrete events; master conservation and reversibility.
 - 知道 $8+4=12$ 所以 $12-4=8$ ，但还不能思考抽象假设性问题
- **Formal operational stage**
 - Piaget's **Stage 4** (~12+): capacity for **abstract, hypothetical, and systematic** thinking.
 - 能理解正义、自由这些抽象概念，能设计科学实验
- **Object permanence**
 - The understanding that objects continue to **exist even when not seen**; develops around 8-10 months.
 - 把玩具藏在毯子下面，婴儿知道去掀开毯子找
- **Mental symbols**
 - Internal representations of objects or events that allow thinking **without direct sensory experience**.
 - 婴儿可以在脑中「想到」一个不在眼前的东西

- **Pretend play**
 - Using objects or actions to **represent** other objects or actions symbolically; emerges in preoperational stage.
 - 小孩拿香蕉当电话打，过家家
- **Conservation**
 - The understanding that quantity remains the same despite changes in **shape or appearance**.
 - 把同样多的水倒进高瘦杯子和矮胖杯子，水量不变
- **Reversibility**
 - The ability to mentally **reverse** an action or operation.
 - 知道 $8+4=12$ ，也就知道 $12-4=8$
- **Animism**
 - The belief that **inanimate objects are alive** and have feelings; common in preoperational stage.
 - 小孩说太阳公公在对我笑，觉得玩具熊会疼
- **Egocentrism**
 - The inability to take **another person's perspective**; characteristic of preoperational stage.
 - 三山实验：小孩觉得别人看到的山和自己看到的一样
- **Theory of mind**
 - The ability to understand that **others have different thoughts, beliefs, and perspectives** than one's own.
 - 知道别人可能不知道你知道的事情，大约在4-5岁发展出来
- **Systematic thinking**
 - The ability to approach problems in an **organized, logical** manner; develops in concrete operational stage.
 - 能有条理地一步步解决问题，而不是乱猜
- **Abstract thinking**
 - The ability to think about concepts that are **not physically present**; develops in formal operational stage.
 - 能思考「正义是什么」「 $y=ax+b$ 里x代表什么」
- **Hypothetical thinking**
 - The ability to consider "**what if**" scenarios; develops in formal operational stage.
 - 如果地球没有引力会怎样？

- **Scaffolding (as it pertains to Vygotsky)**

- Temporary support provided by a more knowledgeable person to help a learner accomplish a task within their **ZPD**.
- 老师先给你提示和框架，等你学会了就慢慢撤掉帮助

- **Zone of proximal development**

- The gap between what a learner can do **independently** and what they can do **with guidance**.
- 你自己解不出这道数学题，但老师提示一下你就会了，这个区间就是ZPD

- **Crystallized intelligence**

- Accumulated **knowledge and verbal skills** that tend to **increase** with age.
- 老人的词汇量、常识、经验比年轻人丰富

- **Fluid intelligence**

- The ability to think quickly, abstractly, and solve novel problems; tends to **decline** with age.
- 年轻人反应快、学新东西快，这种能力随年龄下降

- **Dementia**

- A **progressive decline** in cognitive function (memory, reasoning, language) severe enough to interfere with daily life.
- 老人记忆力严重衰退，忘记家人的名字，无法独立生活

3.5 - Communication and Language Development

- **Phonemes**
 - The **smallest units of sound** that distinguish meaning in a language.
 - /b/ 和 /p/ 是不同的音位，bat 和 pat 意思不同
- **Morphemes**
 - The **smallest units of meaning** in a language.
 - un-believe-able 有三个语素：un表示不，believe表示相信，able表示能够
- **Semantics**
 - The study of **meaning** in language — what words and sentences actually convey.
 - 理解「他很冷」可能是说温度低，也可能是说这个人冷漠
- **Grammar**
 - The **system of rules** governing the structure and use of a language.
 - 英语中主语+谓语+宾语的规则
- **Syntax**
 - Rules for **combining words into grammatically correct sentences**.
 - The dog bit the man 和 The man bit the dog 语序不同意思完全不同
- **Cooing**
 - Vowel-like sounds produced by infants beginning around **2 months**.
 - 婴儿发出「噢噢啊啊」的声音，在练习发声器官
- **Babbling**
 - Repetitive consonant-vowel combinations beginning around **4 months**.
 - ba-ba-ba, ma-ma-ma, 开始练习母语的音位
- **One-word stage**
 - Around **12 months**, children use single words to express whole ideas (holophrases).
 - 指着门说「出！」意思是我想出去玩
- **Telegraphic speech**
 - Two-word sentences (around age **2**) that contain mostly nouns and verbs, like a telegram.
 - 妈妈走、要球球，只有关键词没有语法词
- **Overgeneralization of language rules**
 - Applying grammatical rules **too broadly**, including to exceptions.
 - 孩子说 goed 而不是 went，说 mouses 而不是 mice

3.6 - Social-Emotional Development Across the Lifespan

- **Ecological systems theory**
 - Bronfenbrenner's theory that development is influenced by **multiple nested environmental systems**.
 - 一个孩子的发展受到家庭、学校、社区、文化等多层环境的影响
- **Microsystem**
 - The **immediate** environments a person directly interacts with.
 - 家庭、学校、同伴、老师
- **Mesosystem**
 - The **interactions between** microsystems.
 - 家长和老师沟通孩子的表现，家庭和学校之间的联系
- **Exosystem**
 - External environments that **indirectly** affect the individual.
 - 爸爸公司裁员导致家庭经济紧张，间接影响孩子
- **Macrosystem**
 - The broader **cultural values, laws, and customs** that shape all other systems.
 - 中国文化重视教育，影响了家庭对孩子学习的要求
- **Chronosystem**
 - The dimension of **time** — historical events and life transitions that influence development.
 - 经历新冠疫情的这代孩子的成长与之前的不同
- **Authoritarian parenting**
 - **High demands, low responsiveness**; strict rules with little warmth.
 - 我说什么你就做什么，不许反驳，但也不关心你的感受
- **Authoritative parenting**
 - **High demands, high responsiveness**; firm but warm and communicative. **Best outcomes**.
 - 对你有要求有规矩，但也倾听你、支持你，讲道理
- **Permissive parenting**
 - **Low demands, high responsiveness**; warm but few rules or structure.
 - 宠溺孩子，什么都答应，不设规矩
- **Attachment styles**
 - Patterns of bonding between infant and caregiver, shaped by caregiver **responsiveness**.
 - 婴儿和照顾者之间形成的依恋模式，影响一生的人际关系

- **Secure attachment**
 - Child uses caregiver as a **safe base**; distressed when caregiver leaves, happy when they return.
 - 妈妈在就安心探索，妈妈走了会哭，妈妈回来了很开心，最健康的依恋
- **Insecure attachment**
 - A general category for attachment patterns that are **not secure**, including avoidant, anxious, and disorganized.
 - 不安全的依恋，有几种不同的表现形式
- **Avoidant attachment**
 - Child shows **little distress** when caregiver leaves and avoids them upon return; suppresses emotional needs.
 - 妈妈走了无所谓，回来了也无所谓，内心压抑情感需求
- **Anxious attachment**
 - Child is **clingy** when caregiver is present, extremely distressed when they leave, and **ambivalent** upon return.
 - 妈妈在就黏着，妈妈走了崩溃大哭，妈妈回来了又推开她又要抱
- **Disorganized attachment**
 - Child shows **no consistent pattern** of behavior; often linked to abuse or trauma. **Most unhealthy.**
 - 行为混乱没有规律，一会靠近一会远离，通常和虐待或创伤有关
- **Separation anxiety**
 - Distress displayed by infants when separated from their primary caregiver, typically emerging around **8 months**.
 - 妈妈一离开视线婴儿就哭闹不安
- **Temperament**
 - An infant's **inborn** disposition — easy, difficult, or slow to warm up.
 - 有的宝宝天生好带，有的天生难带，有的慢热
- **Parallel play**
 - Children play **side by side** but do not interact with each other.
 - 两个小孩坐在一起各玩各的积木，互不干涉
- **Imaginary audience**
 - Adolescents' belief that **everyone is watching and judging** them.
 - 长了个痘觉得全世界都在看，觉得自己站在舞台中央
- **Personal fable**
 - Adolescents' belief that they are **unique and invulnerable**.
 - 觉得自己与众不同、刀枪不入，不好的事不会发生在自己身上

- **Social clock**
 - The culturally preferred timing for major life events like marriage, career, and retirement.
 - 社会觉得你该什么年龄结婚、生孩子、退休，带来无形压力
- **Emerging adulthood**
 - The transitional period from late teens to mid-20s, characterized by **identity exploration** and instability.
 - 18到25岁左右，不完全是青少年也不完全是成年人，还在探索
- **Stage theory of psychosocial development (Erikson)**
 - Erikson's **8-stage** theory proposing that personality develops through a series of crises across the lifespan.
 - 每个人生阶段都有一个核心心理冲突要解决
- **Trust and mistrust**
 - **Stage 1** (infancy): Can I trust the world? Consistent caregiving leads to trust.
 - 婴儿期：照顾者靠谱就信任世界，不靠谱就不信任
- **Autonomy and shame and doubt**
 - **Stage 2** (toddlerhood): Can I do things myself? Encouragement leads to autonomy.
 - 幼儿期：鼓励孩子自己做事就有自主感，总批评就会自我怀疑
- **Initiative and guilt**
 - **Stage 3** (preschool): Can I make things happen? Support leads to initiative.
 - 学前期：让孩子主动探索就有主动性，总阻止就会内疚
- **Industry and inferiority**
 - **Stage 4** (school age): Can I succeed? Mastery leads to competence.
 - 学龄期：在学校获得成就感就勤奋，总是失败就自卑
- **Identity and role confusion**
 - **Stage 5** (adolescence): Who am I? Successful exploration leads to a strong sense of identity.
 - 青春期：探索自我，我是谁，我要成为什么样的人
- **Intimacy and isolation**
 - **Stage 6** (young adulthood): Can I love? Forming close relationships vs. feeling alone.
 - 青年期：能不能建立亲密关系，还是孤独一人
- **Generativity and stagnation**
 - **Stage 7** (middle adulthood): Am I contributing to the next generation?
 - 中年期：为下一代和社会做贡献，还是停滞不前觉得人生没意义

- **Integrity and despair**
 - **Stage 8** (late adulthood): Did I live a meaningful life?
 - 老年期：回顾一生觉得满足，还是充满遗憾和绝望
- **Adverse childhood experiences (ACEs)**
 - Traumatic events in childhood (abuse, neglect, household dysfunction) that have **lasting negative effects** on health and development.
 - 童年遭受虐待、忽视、家庭暴力等，对身心健康产生长期负面影响
- **Achievement (adolescent identity development)**
 - Marcia's status: the individual has **explored and committed** to an identity.
 - 经过探索后做出了明确的选择，比如想清楚了要学什么专业
- **Diffusion (adolescent identity development)**
 - Marcia's status: **no exploration, no commitment**.
 - 没想过也不在乎自己是谁、要做什么，随波逐流
- **Foreclosure (adolescent identity development)**
 - Marcia's status: **commitment without exploration**; identity adopted from others (e.g., parents).
 - 爸妈说你要当医生，那就当医生吧，自己从没想过别的可能
- **Moratorium (adolescent identity development)**
 - Marcia's status: **actively exploring but no commitment** yet.
 - 正在尝试各种可能性，还没做出最终决定
- **Racial/ethnic identity**
 - One's sense of belonging to a particular **racial or ethnic group** and how it shapes self-concept.
 - 作为华裔对自己民族文化身份的认同和理解
- **Sexual orientation**
 - An enduring pattern of emotional, romantic, and/or sexual attraction to others.
 - 一个人对同性、异性或两性的持久吸引力倾向
- **Religious identity**
 - One's sense of self as shaped by **religious beliefs and practices**.
 - 信仰佛教或基督教对自我认同的影响
- **Occupational identity**
 - One's sense of self related to **career choices and professional roles**.
 - 我想成为交易员，这是我职业身份的一部分

- **Familial identity**

- One's sense of self shaped by **family roles and relationships**.
- 作为家里的长子，对自己的角色和责任的认知

- **Possible selves**

- Mental representations of what one **hopes to become, expects to become, or fears becoming**.
- 我希望成为的自己、我害怕变成的自己

3.7 - Classical Conditioning

- **Behavioral perspective**
 - The psychological approach that focuses on **observable behavior** and how it is learned from the environment.
 - 只关注能观察到的行为，不深究内心想法
- **Classical conditioning**
 - A type of learning where a **neutral stimulus** becomes associated with an **unconditioned stimulus** to produce a conditioned response.
 - 巴甫洛夫的狗：铃声反复和食物配对后，听到铃声就流口水
- **Association**
 - The mental link formed between two stimuli or between a stimulus and a response.
 - 把铃声和食物联系在一起
- **Acquisition**
 - The initial stage of learning, when a response is **first established** and gradually strengthened.
 - 小猫逐渐学会听到罐头声就跑去厨房
- **Associative learning**
 - Learning that certain events **occur together**; includes both classical and operant conditioning.
 - 通过一个刺激预测另一个刺激的出现
- **Unconditioned stimulus**
 - A stimulus that **naturally and automatically** triggers a response without prior learning.
 - 食物自动引发流口水，打针的疼痛自动引发哭泣
- **Unconditioned response**
 - The **natural, unlearned** reaction to the unconditioned stimulus.
 - 吃到食物自然流口水，被打针自然会哭
- **Conditioned stimulus**
 - A previously **neutral stimulus** that, after association with the US, now **triggers a learned response**.
 - 铃声本来不会让狗流口水，但和食物配对多次后就可以了
- **Conditioned response**
 - The **learned** response to the conditioned stimulus; typically similar to the UR.
 - 听到铃声就流口水，看到白大褂就害怕
- **Extinction**
 - The diminishing of a conditioned response when the CS is **repeatedly presented without the US**.
 - 铃声响了很多次但没有食物，狗慢慢不再流口水了

- **Spontaneous recovery**
 - The **reappearance** of an extinguished conditioned response after a rest period.
 - 消退一段时间后，铃声再响，狗又流口水了
- **Stimulus discrimination**
 - The ability to **distinguish** between the CS and other similar stimuli.
 - 小猫只对特定罐头声有反应，对其他类似声音不反应
- **Stimulus generalization**
 - The tendency to respond to stimuli that are **similar** to the CS.
 - 小猫听到任何类似罐头的声音都跑去厨房
- **Higher-order conditioning**
 - A procedure where a **CS is used as a US** to condition a new neutral stimulus.
 - 白大褂让你害怕，白大褂总和某段音乐一起出现，后来听到那段音乐也害怕了
- **Counterconditioning**
 - Replacing an unwanted CR with a new, **incompatible** response to the same CS.
 - 怕狗的人通过在狗旁边吃最喜欢的零食，慢慢把恐惧替换成愉悦
- **Taste aversion**
 - A learned avoidance of a specific food after it was followed by **illness**; often requires **only one trial**.
 - 生病时吃了豆花，从此再也不想吃豆花了
- **One-trial conditioning**
 - Classical conditioning that occurs after **a single** pairing of the CS and US.
 - 只被狗咬了一次就从此怕狗
- **One-trial learning**
 - Learning that occurs from **a single experience**, often involving strong emotional or biological significance.
 - 摸了一次火炉被烫到，再也不敢摸了
- **Biological preparedness**
 - An evolutionary predisposition to more easily learn associations that are **survival-relevant**.
 - 人天生更容易怕蛇和蜘蛛，而不是怕花朵，因为前者对生存有威胁
- **Habituation**
 - **Decreased responsiveness** to a repeated, insignificant stimulus; a CNS-level learned process, **not** sensory adaptation.
 - 住在铁路旁边的人慢慢不再注意到火车的声音了

3.8 - Operant Conditioning

- **Operant conditioning**
 - A type of learning where behavior is strengthened or weakened by its **consequences** (reinforcement or punishment).
 - 行为的后果决定这个行为以后会不会再做
- **Reinforcement**
 - Any consequence that **increases** the likelihood of a behavior being repeated.
 - 任何让你更想重复某个行为的后果
- **Punishment**
 - Any consequence that **decreases** the likelihood of a behavior being repeated.
 - 任何让你不想再做某个行为的后果
- **Law of Effect**
 - Thorndike's principle: behaviors followed by **satisfying** consequences are more likely to recur; those followed by **unpleasant** consequences are less likely.
 - 效果好的行为会重复，效果差的行为会减少
- **Superstitious behavior**
 - A behavior that is repeated because a reinforcer **coincidentally** followed it, even though there is no causal relationship.
 - 考试穿了某件衣服考得好，以后每次考试都穿那件
- **Learned helplessness**
 - A condition where repeated exposure to unavoidable negative events leads to **passivity and giving up**.
 - 反正怎么努力都没用，干脆摆烂算了
- **Positive reinforcement**
 - **Adding** a pleasant stimulus after a behavior to **increase** that behavior. 加好。
 - 考试考好了给奖金，工作表现好发奖金
- **Primary reinforcers**
 - Stimuli that are **innately** reinforcing, satisfying biological needs.
 - 食物、水、睡眠，不需要学习就知道它们好
- **Secondary reinforcer**
 - Stimuli that become reinforcing through **learned association** with primary reinforcers.
 - 钱、分数、小红花，本身没意义但可以换成食物或认可

- **Reinforcement discrimination**
 - Responding differently to stimuli that **signal** whether reinforcement is available.
 - 只在老板在的时候认真工作，因为知道老板在才有可能被表扬
- **Reinforcement generalization**
 - Responding similarly to stimuli that are **similar** to those previously associated with reinforcement.
 - 在一个老师那里认真学习得到表扬，在其他老师那里也认真
- **Shaping**
 - Reinforcing **successive approximations** toward a desired behavior, step by step.
 - 教狗翻滚：先奖励它躺下，再奖励它侧翻，最后奖励它完整翻滚
- **Instinctive drift**
 - The tendency for an animal to **revert to innate behaviors** that interfere with conditioned behavior.
 - 训练猪把硬币放进存钱罐，但猪总是忍不住用鼻子拱硬币，因为这是本能
- **Negative reinforcement**
 - **Removing** an unpleasant stimulus after a behavior to **increase** that behavior. 减坏。
 - 系安全带后烦人的警报声停了，以后上车就系安全带
- **Reinforcement schedules**
 - Rules that determine **when and how often** a behavior is reinforced.
 - 决定什么时候给强化的规则
- **Continuous reinforcement**
 - Reinforcing the behavior **every time** it occurs. Fastest learning, fastest extinction.
 - 每次按按钮都出饮料，一次不出就不按了
- **Partial reinforcement**
 - Reinforcing behavior only **some of the time**. Slower learning, but much more resistant to extinction.
 - 老虎机不是每次都中奖，但人们还是不停地玩
- **Fixed interval**
 - Reinforcement after a **fixed time period**; response rate increases as the time approaches.
 - 按月发工资，月底前才开始努力工作；考前突击
- **Variable interval**
 - Reinforcement after **unpredictable time periods**; produces steady, consistent responding.
 - 随堂测验不知道什么时候考，所以每天都要准备；刷社交媒体等点赞

- **Fixed ratio**

- Reinforcement after a **fixed number** of responses; produces high rate with brief pauses after reinforcement.
- 计件工资，每做50个产品得20块；买十杯咖啡送一杯

- **Variable ratio**

- Reinforcement after an **unpredictable number** of responses; produces the **highest and most consistent** response rate.
- 老虎机、抽卡游戏，不知道什么时候中但就是停不下来

- **Scalloped graph**

- The response pattern seen in **fixed interval** schedules — low responding after reinforcement, then a sharp increase as the next reinforcement approaches.
- 固定间隔强化的图表长得像扇贝壳，刚拿完工资就摸鱼，快发工资了才干活

3.9 - Social, Cognitive, and Neurological Factors in Learning

- **Social learning theory**
 - Bandura's theory that we learn by **observing and imitating** others, not just through direct experience.
 - 小孩看到大人打充气娃娃，自己也跟着打（Bobo doll实验）
- **Vicarious conditioning**
 - Learning by observing the **consequences of others'** behavior.
 - 看到同学上课说话被罚站，自己就不敢说话了
- **Modeling**
 - The process of **observing and imitating** a specific behavior.
 - 小孩模仿爸爸刮胡子的动作
- **Insight learning**
 - A sudden realization of a problem's solution, an **"aha" moment**, without trial and error.
 - 想了很久的数学题突然灵光一闪就会了
- **Latent learning**
 - Learning that occurs **without reinforcement** and is not demonstrated until there is a reason to do so.
 - 你每天坐车经过同一条路，虽然没刻意记，但有一天需要自己走就发现其实认识路
- **Cognitive maps**
 - Mental representations of the **layout of an environment**.
 - 你脑中有一张学校的地图，知道哪里是食堂哪里是教室

4.1 - Attribution Theory and Person Perception

- | | | |
|--|--|---|
| ☐ Attributions | ☐ Actor/observer bias | ☐ Self-fulfilling prophecy |
| ☐ Dispositional attributions | ☐ Fundamental attribution error | ☐ Social comparison |
| ☐ Situational attributions | ☐ Self-serving bias | ☐ Upward social comparison |
| ☐ Explanatory style | ☐ Internal locus of control | ☐ Downward social comparison |
| ☐ Optimistic explanatory style | ☐ External locus of control | ☐ Relative deprivation |
| ☐ Pessimistic explanatory style | ☐ Mere exposure effect | |

4.2 - Attitude Formation and Attitude Change

- | | | |
|---|---|---|
| ☐ Stereotype | ☐ Ethnocentrism | ☐ Belief perseverance |
| ☐ Prejudice | ☐ Just-world phenomenon | ☐ Confirmation bias |
| ☐ Discrimination | ☐ Out-group homogeneity bias | ☐ Cognitive dissonance |
| ☐ Implicit attitudes | ☐ In-group bias | ☐ Cognitive load |

4.3 - Psychology of Social Situations

- | | | |
|---|--|--|
| ☐ Social norms | ☐ Obedience | ☐ Social traps |
| ☐ Social influence theory | ☐ Individualism | ☐ Industrial-organizational (I/O) psychologists |
| ☐ Normative social influence | ☐ Collectivism | ☐ Burnout |
| ☐ Informational social influence | ☐ Multiculturalism | ☐ Altruism |
| ☐ Persuasion | ☐ Group polarization | ☐ Prosocial behavior |
| ☐ Elaboration likelihood model | ☐ Groupthink | ☐ Social debt |
| ☐ Central route | ☐ Diffusion of responsibility | ☐ Social reciprocity norm |
| ☐ Peripheral route | ☐ Social loafing | ☐ Social responsibility norm |
| ☐ Halo effect | ☐ Deindividuation | ☐ Bystander effect |
| ☐ Foot-in-the-door technique | ☐ Social facilitation | ☐ Situational variables |
| ☐ Door-in-the-face technique | ☐ False consensus effect | ☐ Attentional variables |
| ☐ Conformity | ☐ Superordinate goals | |

4.4 - Psychodynamic and Humanistic Theories of Personality

- | | | |
|---|---|--|
| ☐ Psychodynamic theory | ☐ Rationalization | ☐ Preconscious mind |
| ☐ Unconscious processes | ☐ Reaction formation | ☐ Unconscious mind |
| ☐ Ego defense mechanisms | ☐ Regression | ☐ Humanistic psychology |
| ☐ Denial | ☐ Repression | ☐ Unconditional regard |
| ☐ Displacement | ☐ Sublimation | ☐ Self-actualizing tendency |
| ☐ Projection | ☐ Projective tests | |

4.5 - Social-Cognitive and Trait Theories of Personality

- | | | |
|--|--|--|
| ☐ Social-cognitive theory | ☐ Trait theories | ☐ Openness to experience |
| ☐ Reciprocal determinism | ☐ Big Five theory | ☐ Emotional Stability |
| ☐ Self-concept | ☐ Agreeableness | ☐ Personality inventories |
| ☐ Self-efficacy | ☐ Extraversion | ☐ Factor analysis |
| ☐ Self-esteem | ☐ Conscientiousness | |

4.6 - Motivation

- | | | |
|--|--|---|
| ☐ Drive-reduction theory | ☐ Extrinsic motivation | ☐ Sensation-seeking theory |
| ☐ Homeostasis | ☐ Incentive theory | ☐ Thrill seeking |
| ☐ Arousal theory | ☐ Instincts | ☐ Adventure seeking |
| ☐ Optimal level of arousal | ☐ Lewin's motivational conflicts theory | ☐ Disinhibition |
| ☐ Yerkes-Dodson Law | ☐ Approach-approach conflicts | ☐ Boredom susceptibility |
| ☐ Self-determination theory | ☐ Approach-avoidance conflicts | ☐ Hunger |
| ☐ Intrinsic motivation | ☐ Avoidance-avoidance conflicts | ☐ Satiety |

4.7 - Emotion

- | | | |
|--|--|--|
| ☐ Affect | ☐ Cognitive label | ☐ Cognitive appraisal |
| ☐ Physiological experiences | ☐ Facial-feedback hypothesis | ☐ Display rules |
| ☐ Cognitive experiences | ☐ Broaden-and-build theory of emotion | ☐ Emotional elicitors |

4.1 - Attribution Theory and Person Perception

- **Attributions**

- How we explain the causes of events or behaviors — either internal or external.
- 我们怎么解释一件事发生的原因，比如他为什么迟到

- **Dispositional attributions**

- Explaining behavior as caused by **internal** factors like personality, ability, or effort.
- 他考试考得好，是因为他聪明

- **Situational attributions**

- Explaining behavior as caused by **external** factors like luck, context, or difficulty.
- 他考试考得好，是因为这次题目简单

- **Explanatory style**

- A person's habitual way of explaining events, either optimistically or pessimistically.
- 一个人习惯性地怎么解释发生在自己身上的事

- **Optimistic explanatory style**

- Bad events are seen as **temporary, external, and specific**.
- 考试考砸了：这次题太偏了，下次不会这样

- **Pessimistic explanatory style**

- Bad events are seen as **permanent, internal, and global**.
- 考试考砸了：我就是笨，什么都学不好

- **Actor/observer bias**

- When we are the **actor**, we blame the situation; when we are the **observer**, we blame the person's character. It highlights the **double standard** between judging self vs. others.
- 自己迟到是因为堵车，别人迟到是因为没有时间观念

- **Fundamental attribution error**

- The tendency to **overestimate dispositional factors** and underestimate situational factors **when judging others**.
- 看到别人考试挂了，觉得他就是不努力，而不考虑他可能生病了

- **Self-serving bias**

- Attributing **success to internal** factors and **failure to external** factors to protect self-esteem.
- AP考了5分：我是天才；考了2分：题出得太偏了

- **Internal locus of control**

- Belief that you **control your own outcomes** through effort and decisions.
- 考前认真刷题，相信努力能改变结果

- **External locus of control**
 - Belief that outcomes are determined by **luck, fate, or others**. Can lead to *learned helplessness*.
 - 考前转发锦鲤，觉得考多少分是命
- **Mere exposure effect**
 - Simply being exposed to something repeatedly increases our **liking** for it.
 - 一首歌刚听觉得一般，循环十遍后觉得真上头；广告反复轰炸也是这个原理
- **Self-fulfilling prophecy**
 - A belief leads to behavior that makes the belief come true: **belief → behavior → outcome**.
 - 老师觉得Leo是好学生，给他更多关注，Leo更努力，最终真的成了好学生
- **Social comparison**
 - Evaluating ourselves by comparing with others.
 - 看看身边同学考了多少分来判断自己考得好不好
- **Upward social comparison**
 - Comparing with someone **better** — can motivate or cause feelings of inadequacy.
 - 和年级第一比，觉得自己还要更努力，或者觉得自卑
- **Downward social comparison**
 - Comparing with someone **worse** — boosts self-esteem but may reduce motivation.
 - 和倒数第一比，觉得自己还不错嘛
- **Relative deprivation**
 - Feeling deprived or dissatisfied **after comparing** with others who have more.
 - 本来对工资很满意，知道同事比自己多两千后就不开心了

4.2 - Attitude Formation and Attitude Change

- **Stereotype**
 - An **overgeneralized belief** about a group of people; a cognitive shortcut (heuristic). Not necessarily negative but ignores individual differences.
 - 亚洲人都擅长数学，法国人都很浪漫
- **Prejudice**
 - An **unjustified negative attitude** toward a group — this is the **emotional** level.
 - 我就是不喜欢吃榴莲的人
- **Discrimination**
 - **Unjust behavior** toward a group — this is the **action** level, often caused by prejudice.
 - 因为不喜欢吃榴莲的人，所以不让他们加入我的小组
- **Implicit attitudes**
 - Attitudes that are **unconscious** and automatically activated; people may not be aware they hold them.
 - 嘴上说不歧视，但在无意识的测试中表现出对某群体的偏见（如IAT内隐联想测试）
- **Ethnocentrism**
 - Believing your own culture is **superior** and judging other cultures by your own standards.
 - 看到别人用手吃饭就觉得脏、落后，而不是理解这是不同的文化习俗
- **Just-world phenomenon**
 - The belief that people **get what they deserve**, which often leads to **blaming the victim**.
 - 看到路边的乞丐就想：他肯定是年轻时不努力才沦落到这样
- **Out-group homogeneity bias**
 - Perceiving members of an **out-group as all alike**, while seeing your own group as diverse.
 - 你能分清重金属、朋克、独立摇滚的区别，但觉得K-pop的歌听起来全是一个调子
- **In-group bias**
 - Favoring members of your **own group** over out-group members.
 - 平时吐槽自己学校，但跟外校比赛时觉得我校最棒
- **Belief perseverance**
 - Clinging to initial beliefs **even after they have been discredited** by evidence.
 - 阴谋论者看到科学证据证明地球是圆的，说这都是伪造的，地球就是平的
- **Confirmation bias**
 - Seeking information that **supports** existing beliefs and ignoring contradictory evidence.
 - 觉得每次洗车后都下雨，是因为只记住了洗车后下雨的次数，忘了没下雨的

- **Cognitive dissonance**

- The psychological **discomfort** when attitudes and behaviors are **inconsistent**. To reduce it, we change either the attitude or the behavior.
- 知道抽烟有害还继续抽，于是说服自己：抽烟能减压嘛。吃不到葡萄说葡萄酸

- **Cognitive load**

- The amount of **mental effort** being used in working memory. High cognitive load makes it harder to think critically and easier to rely on heuristics.
- 一边开车一边打电话，大脑超负荷，容易出错

4.3 - Psychology of Social Situations

- **Social norms**
 - Unwritten rules about how to behave in a group or society.
 - 在图书馆要安静，电梯里面朝门站
- **Social influence theory**
 - The theory that our behavior is influenced by others through **normative** or **informational** pathways.
 - 我们的行为受到身边人的影响，不管是为了合群还是为了获取信息
- **Normative social influence**
 - Conforming to be **accepted** by the group, even if you know they're wrong.
 - 一群朋友都抽烟，你明知不好但为了合群也跟着抽
- **Informational social influence**
 - Conforming because you believe others have **better information** than you.
 - 第一次吃西餐不知道怎么用刀叉，偷看隔壁桌怎么做
- **Persuasion**
 - The process of changing someone's attitudes or behaviors through communication.
 - 广告、演讲、谈判中试图改变你的想法
- **Elaboration likelihood model**
 - A model explaining two routes of persuasion: **central** (logic) and **peripheral** (surface cues).
 - 我们被说服有两条路径：走脑子或走感觉
- **Central route**
 - Persuasion through **logic, evidence, and careful thinking**. Leads to **lasting** attitude change.
 - 买电脑时仔细对比CPU、内存和价格
- **Peripheral route**
 - Persuasion through **surface cues** like attractiveness or celebrity endorsement. Effect is **temporary**.
 - 我家哥哥代言联想电脑，买！
- **Halo effect**
 - The tendency to let one positive trait (often **attractiveness**) influence overall judgment of a person.
 - 长得好看的人，别人会觉得他性格也好、能力也强
- **Foot-in-the-door technique**
 - Getting someone to agree to a **small request first**, then following with a larger one.
 - 先请你签名支持环保，然后问你能不能捐100块

- **Door-in-the-face technique**
 - Starting with a **large request** that will be refused, then following with a smaller one.
 - 先问借5000买iPad，被拒后说那借100吃个饭总行吧
- **Conformity**
 - Adjusting behavior or thinking to match **group standards**.
 - 电梯里所有人都面朝后站，你不自觉也转过身去
- **Obedience**
 - Complying with direct commands from an **authority figure**. (Conformity = peers; Obedience = authority)
 - 警察说不许动，你就不动了
- **Individualism**
 - Cultural emphasis on **personal goals, independence, and self-expression**.
 - 美国文化强调个人自由和独立思考
- **Collectivism**
 - Cultural emphasis on **group harmony, duty, and interdependence**.
 - 东亚文化强调家庭和集体利益
- **Multiculturalism**
 - Valuing and integrating **diverse cultural perspectives** within a society.
 - 一个社区里不同文化背景的人互相尊重、和平共处
- **Group polarization**
 - Group discussion makes members' views become **more extreme** than their initial positions.
 - 一群观点相似的人讨论后，观点变得更加极端，比如越聊越激动地吐槽某个人
- **Groupthink**
 - Desire for group harmony suppresses **dissenting opinions**, leading to poor decisions.
 - 开会时大家都觉得方案有问题，但没人敢说，结果项目失败了
- **Diffusion of responsibility**
 - The more people present, the **less personal responsibility** each person feels to act.
 - 有人倒在路上，围观的人越多，每个人越觉得别人会去帮忙
- **Social loafing**
 - Individuals exert **less effort** when working in a group because their contribution can't be individually measured.
 - 三个和尚没水吃，小组作业总有人摸鱼

- **Deindividuation**
 - Loss of self-awareness and self-restraint in group situations due to **anonymity and arousal**.
 - 演唱会上疯狂尖叫跟唱，键盘侠在网上骂人
- **Social facilitation**
 - Presence of others **improves** performance on simple/well-practiced tasks but **impairs** performance on complex/new tasks.
 - 台球高手有人看打得更准，但第一次演讲有人看反而忘词
- **False consensus effect**
 - Overestimating how much others **agree** with our own beliefs or behaviors.
 - 你觉得香菜奶茶超好喝，以为大家都喜欢，买了一堆请全班结果没人喝
- **Superordinate goals**
 - A shared goal that requires **cooperation** between conflicting groups, helping to reduce hostility.
 - 两个互相仇视的小队必须一起推出陷在泥里的卡车，推完关系就好了
- **Social traps**
 - Individuals pursuing **short-term self-interest** that leads to **long-term collective harm**.
 - 大家都开私家车图方便，结果导致严重拥堵和雾霾
- **Industrial-organizational (I/O) psychologists**
 - Psychologists who apply psychological principles to the **workplace** to improve productivity and employee well-being.
 - 公司请心理学家设计面试题，或优化办公室布局让员工更高效
- **Burnout**
 - Physical and emotional **exhaustion** caused by prolonged work-related stress.
 - 连续加班几个月后感到身心俱疲，对工作完全失去热情
- **Altruism**
 - Selfless concern for the well-being of others, with **no expectation of reward**.
 - 匿名捐款给灾区，不求任何回报
- **Prosocial behavior**
 - Any action intended to **help others**, which may or may not be selfless.
 - 帮老奶奶过马路，扶起摔倒的小朋友
- **Social debt**
 - The feeling of **owing** someone after they have helped you.
 - 朋友帮你搬家后，你觉得欠了个人情

- **Social reciprocity norm**
 - The expectation that people will **return favors** — help those who have helped you.
 - 别人帮了我，我以后也要帮他
- **Social responsibility norm**
 - The expectation that people should **help those in need**, even if they can't repay.
 - 帮助弱势群体，即使他们无法回报
- **Bystander effect**
 - The more people present at an emergency, the **less likely** any individual is to help.
 - 大街上有人晕倒，围观的人很多但没人打120
- **Situational variables**
 - External factors in the environment that influence whether someone will help.
 - 天黑、偏僻的地方，人们更不愿意伸出援手
- **Attentional variables**
 - Whether a person **notices** the event and recognizes it as an emergency, affecting helping behavior.
 - 戴着耳机走路，根本没注意到有人摔倒了

4.4 - Psychodynamic and Humanistic Theories of Personality

- **Psychodynamic theory**

- Freud's approach emphasizing **unconscious processes, childhood experiences, and internal conflicts** as drivers of personality and behavior.
- 弗洛伊德认为人的行为受潜意识驱动，童年经历影响一生

- **Unconscious processes**

- Thoughts, desires, and memories that are outside conscious awareness but still influence behavior. The deepest level of Freud's iceberg model.
- 被压抑的童年创伤记忆，自己想不起来但影响着现在的行为

- **Ego defense mechanisms**

- Unconscious strategies the **ego** uses to reduce anxiety from conflicts between the id, superego, and reality.
- 大脑自动启动的心理保护机制，用来缓解焦虑

- **Denial**

- Refusing to accept a painful reality.
- 亲人去世后说：这不可能，一定是搞错了！

- **Displacement**

- Redirecting emotions toward a **safer or weaker target**.
- 被老板骂了不敢还嘴，回家对家人撒气

- **Projection**

- Attributing your own unacceptable thoughts or feelings to **someone else**.
- 以小人之心度君子之腹，自己自私却觉得别人都自私

- **Rationalization**

- Creating logical-sounding excuses to justify unacceptable behavior or feelings.
- 考砸了说不是没努力，是考试太难了

- **Reaction formation**

- Behaving in a way **opposite** to one's true feelings.
- 喜欢一个人却故意对他冷淡甚至刻薄

- **Regression**

- Reverting to **childlike** behaviors when stressed.
- 成年人遇到压力大哭大闹要找妈妈

- **Repression**
 - Unconsciously **blocking** disturbing thoughts or memories from awareness. (The most basic defense mechanism)
 - 完全想不起小时候被伤害的经历，不是故意忘而是真的记不起来
- **Sublimation**
 - Channeling unacceptable impulses into **socially acceptable** or productive activities.
 - 化悲愤为力量，失恋后疯狂健身
- **Projective tests**
 - Personality assessments using **ambiguous stimuli** (e.g., inkblots) to reveal unconscious thoughts. Limited in *reliability and validity*.
 - 罗夏墨迹测验：给你看一团墨迹，让你说看到了什么，以此推断潜意识
- **Preconscious mind**
 - Information not currently in awareness but **easily retrievable**.
 - 你现在没在想你早餐吃了什么，但一问你马上就能想起来
- **Unconscious mind**
 - The deepest level containing **repressed** thoughts, desires, and memories that are not accessible to conscious awareness.
 - 被压抑的痛苦记忆和不可接受的欲望，自己根本意识不到
- **Humanistic psychology**
 - An approach emphasizing **free will, self-actualization, and inherent goodness**. Key figure: Carl Rogers.
 - 每个人都有积极向上的潜能，强调人的自由意志
- **Unconditional regard**
 - Accepting and loving someone **without judgment or conditions**. (Unconditional Positive Regard)
 - 无论孩子做了什么，父母都表达接纳和爱，不附带条件
- **Self-actualizing tendency**
 - The innate drive to **grow, develop, and fulfill one's potential**.
 - 人天生就有不断追求成为更好自己的内在驱动力

4.5 - Social-Cognitive and Trait Theories of Personality

- **Social-cognitive theory**

- Personality is shaped by the **interaction** of personal cognitive factors, behavior, and environment.
- 人的性格是认知、行为和环境三者相互影响的结果

- **Reciprocal determinism**

- Bandura's concept that **personal factors, behavior, and environment** all mutually influence each other.
- 一个被贴上坏学生标签的孩子（环境）→ 自我认知变差（认知）→ 不想学习（行为）→ 成绩更差（环境强化）→ 恶性循环

- **Self-concept**

- A person's overall perception and belief about **who they are**. (Answers "Who am I?")
- 我是一个内向的、喜欢画画的高中生

- **Self-efficacy**

- Belief in one's ability to **succeed at a specific task**.
- 我相信我能通过明天的AP考试

- **Self-esteem**

- One's overall **sense of self-worth**. Unlike self-efficacy, it's about general value, not task-specific.
- 无论考好考差，我都觉得自己是有价值的人

- **Trait theories**

- Theories that describe personality as a set of **stable, measurable characteristics** (traits).
- 用一组稳定的特质来描述人格，比如外向、尽责

- **Big Five theory**

- The most widely accepted trait model: **OCEAN** — Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism (Emotional Stability).
- 用五个维度描述人格的模型

- **Agreeableness**

- Tendency toward trust, altruism, kindness, and cooperation. Low = competitive or manipulative.
- 高分：主动帮助同事；低分：为达目的不择手段

- **Extraversion**

- Tendency toward sociability, talkativeness, and positive emotions. Low (introversion) = reserved, needs alone time.
- 高分：参加派对很兴奋；低分：独处才能充电

- **Conscientiousness**
 - Tendency toward being organized, dependable, and self-disciplined. Low = careless, procrastinating.
 - 高分：每天列待办清单；低分：经常忘记截止日期
- **Openness to experience**
 - Tendency toward curiosity, creativity, and appreciation of new ideas. Low = conventional, prefers routine.
 - 高分：喜欢探索艺术和哲学；低分：喜欢按部就班
- **Emotional Stability**
 - The opposite end of **Neuroticism**. High stability = calm, resilient; Low stability (high neuroticism) = anxious, moody.
 - 高分：泰山崩于前而色不变；低分：遇到小事就焦虑
- **Personality inventories**
 - Standardized **self-report questionnaires** designed to assess personality traits.
 - MMPI、大五人格问卷，通过一系列问题来评估你的人格特质
- **Factor analysis**
 - A **statistical method** used to identify clusters of correlated traits, which was used to develop the Big Five.
 - 通过统计方法把大量人格数据归纳成几个核心因素，大五人格就是这么来的

4.6 - Motivation

- **Drive-reduction theory**
 - Physiological needs create a state of tension (**drive**) that motivates behavior to reduce the drive and restore **homeostasis**.
 - 口渴（需要）→ 想喝水（驱力）→ 喝水（行为）→ 不渴了（平衡恢复）
- **Homeostasis**
 - The body's tendency to maintain a **stable internal state** (e.g., temperature, blood sugar).
 - 身体自动维持体温在37度左右
- **Arousal theory**
 - We are motivated to maintain an **optimal level of arousal** — not too high, not too low.
 - 无聊时想找点刺激，太紧张时想放松
- **Optimal level of arousal**
 - The ideal amount of physiological activation for **best performance**, which varies by person and task.
 - 考试时适度紧张发挥最好，太放松或太焦虑都不行
- **Yerkes-Dodson Law**
 - Performance increases with arousal up to a point; **easy tasks** benefit from higher arousal, **difficult tasks** benefit from lower arousal.
 - 简单任务：来点压力表现更好；困难任务：太紧张反而失常
- **Self-determination theory**
 - Motivation is driven by three innate needs: **autonomy, competence, and relatedness**.
 - 人需要自主感、胜任感和归属感才能保持内在动力
- **Intrinsic motivation**
 - Doing something because it is inherently **interesting or enjoyable**.
 - 因为喜欢画画而画画，不是为了获奖
- **Extrinsic motivation**
 - Doing something for **external rewards** (money, grades, praise).
 - 为了奖金完成销售目标
- **Incentive theory**
 - Behavior is motivated by **external stimuli** (incentives) that pull us toward them.
 - 看到美食图片就产生食欲，闻到咖啡香就想喝
- **Instincts**
 - Innate, **fixed patterns** of behavior that are biologically driven (evolutionary perspective).
 - 婴儿天生会吮吸，母亲保护孩子的本能

- **Lewin's motivational conflicts theory**
 - Kurt Lewin's framework describing three types of conflicts when choosing between goals.
 - 做选择时内心的纠结可以分为三种类型
- **Approach-approach conflicts**
 - Choosing between **two equally attractive** options.
 - 清华还是北大，两个都想去
- **Approach-avoidance conflicts**
 - A single goal has both **attractive and unattractive** aspects.
 - 想吃甜食但又怕胖
- **Avoidance-avoidance conflicts**
 - Choosing between **two equally undesirable** options.
 - 补作业还是请家长，两个都不想
- **Sensation-seeking theory**
 - People differ in their need for **novel, complex, and intense** stimulation.
 - 有些人天生就爱冒险、追求刺激
- **Thrill seeking**
 - Seeking exciting experiences that often involve **physical risk**.
 - 跳伞、蹦极、极限运动
- **Adventure seeking**
 - Desire for **novel physical activities** and exploration.
 - 喜欢旅行探险、尝试新的户外运动
- **Disinhibition**
 - Seeking social stimulation through behaviors that are normally **restrained by social norms**.
 - 在派对上放飞自我，做平时不敢做的事
- **Boredom susceptibility**
 - Intolerance of **repetitive, monotonous, or predictable** situations.
 - 稳定但无聊的工作干不下去，非要换个刺激的
- **Hunger**
 - A motivational state driven by physiological signals (e.g., **ghrelin** from empty stomach, **hypothalamus**) and psychological/environmental cues.
 - 肚子咕咕叫是因为胃分泌了饥饿素，到了饭点也会条件反射地饿

- **Satiety**

- The feeling of **fullness** that signals the body to stop eating, regulated by hormones like **leptin**.
- 吃饱了大脑收到瘦素信号，告诉你别吃了

4.7 - Emotion

- **Affect**
 - A broad term for the experience of **feeling or emotion**.
 - 泛指所有的情绪情感体验
- **Physiological experiences**
 - The **bodily responses** associated with emotion, such as increased heart rate, sweating, or muscle tension.
 - 紧张时心跳加速、手心出汗
- **Cognitive experiences**
 - The **thoughts and interpretations** that accompany an emotional state.
 - 看到蛇时脑子里想到：我在危险中
- **Cognitive label**
 - The **interpretation** we assign to physiological arousal, which determines the emotion we feel. Key to **Schachter-Singer Two-Factor Theory**.
 - 同样是心跳加速，在考场上解读为焦虑，在过山车上解读为兴奋
- **Facial-feedback hypothesis**
 - Facial muscle movements can **trigger or intensify** the corresponding emotion.
 - 即使心情一般，强迫自己微笑也会让你感觉好一点
- **Broaden-and-build theory of emotion**
 - **Positive emotions** broaden our awareness and encourage building new skills, resources, and social connections.
 - 快乐时更愿意探索新事物、交新朋友，积累更多资源
- **Cognitive appraisal**
 - The process of **evaluating an event's significance** to determine the emotional response. Central to **Lazarus's theory**.
 - 你把考试定义为威胁还是挑战，决定了你后续是焦虑还是兴奋
- **Display rules**
 - Cultural norms dictating **when, where, and how** emotions should be expressed.
 - 有些文化提倡喜形于色，有些文化觉得公开哭泣是不成熟的
- **Emotional elicitors**
 - Stimuli or events that **trigger** emotional responses, influenced by gender, age, and culture.
 - 蛇是恐惧的诱因，收到礼物是快乐的诱因

5.1 - Introduction to Health Psychology

- | | | |
|--|---|--|
| ☐ Health psychology | ☐ Distress (debilitating) | ☐ Flight-flight-freeze response |
| ☐ Stress | ☐ Adverse childhood experiences (ACEs) | ☐ Exhaustion phase |
| ☐ Hypertension | ☐ General adaptation syndrome (GAS) | ☐ Tend-and-befriend theory |
| ☐ Immune suppression | ☐ Alarm reaction phase | ☐ Problem-focused coping |
| ☐ Stressors | ☐ Resistance phase | ☐ Emotion-focused coping |
| ☐ Eustress (motivating) | | |

5.2 - Positive Psychology

- | | | |
|--|--|--|
| ☐ Positive psychology | ☐ Positive subjective experiences | ☐ Happiness |
| ☐ Well-being | ☐ Subjective well-being | ☐ Subjective well-being |
| ☐ Resilience | ☐ Signature strengths | ☐ Categories of virtues |
| ☐ Positive emotions | ☐ Virtues | ☐ Posttraumatic growth |
| ☐ Gratitude | ☐ Positive objective experiences | |

5.3 - Explaining and Classifying Psychological Disorders

- | | | |
|---|--|--|
| ☐ Dysfunction | ☐ Discrimination | ☐ Cognitive perspective |
| ☐ Distress | ☐ Diagnostic and Statistical Manual of Mental Disorders (DSM) | ☐ Psychodynamic perspective |
| ☐ Deviation from the social norm | ☐ International Classification of Mental Disorders (ICD) | ☐ Evolutionary perspective |
| ☐ Cultural/societal norms | ☐ Eclectic approach | ☐ Sociocultural perspective |
| ☐ Stigma | ☐ Behavioral perspective | ☐ Biological perspective |
| ☐ Racism | ☐ Maladaptive learned associations | ☐ Biopsychosocial model |
| ☐ Sexism | ☐ Humanistic perspective | ☐ Diathesis-stress model |
| ☐ Ageism | | |

5.4 - Selection of Categories of Psychological Disorders

- | | | |
|--|---|--|
| ☐ Neurodevelopmental disorders | ☐ Mania | ☐ Trauma and stressor-related disorders |
| ☐ Attention-deficit / hyperactivity disorder (ADHD) | ☐ Depression | ☐ Hypervigilance |
| ☐ Autism Spectrum Disorder (ASD) | ☐ Bipolar I Disorder | ☐ Flashbacks |
| ☐ Schizophrenic Spectrum Disorders | ☐ Bipolar II Disorder | ☐ Insomnia |
| ☐ Delusions | ☐ Anxiety disorders | ☐ Emotional detachment |
| ☐ Hallucinations | ☐ Specific phobia | ☐ Hostility |
| ☐ Disorganized thinking | ☐ Acrophobia | ☐ Posttraumatic stress disorder |
| ☐ Disorganized speech | ☐ Arachnophobia | ☐ Feeding and eating disorders |
| ☐ Disorganized motor behavior | ☐ Agoraphobia | ☐ Anorexia nervosa |
| ☐ Negative symptoms | ☐ Panic disorder | ☐ Bulimia nervosa |
| ☐ Positive symptoms | ☐ Panic attack | ☐ Personality disorders |
| ☐ Delusions | ☐ Ataque de nervios | ☐ Cluster A (odd or eccentric cluster) |
| ☐ Delusions of persecution | ☐ Social anxiety disorder | ☐ Paranoid personality disorder |
| ☐ Delusions of grandeur | ☐ Taijin kyofusho | ☐ Schizoid personality disorder |
| ☐ Hallucinations | ☐ Generalized anxiety disorder (GAD) | ☐ Schizotypal personality disorder |
| ☐ Word salad | ☐ Obsessive-compulsive and related disorders | ☐ Cluster B (dramatic, emotional, or erratic cluster) |
| ☐ Catatonia | ☐ Obsessions | ☐ Antisocial personality disorder |
| ☐ Flat affect | ☐ Compulsions | ☐ Histrionic personality disorder |
| ☐ Catatonic stupor | ☐ Obsessive-compulsive disorder | ☐ Narcissistic personality disorder |
| ☐ Dopamine hypothesis | ☐ Hoarding disorder | ☐ Borderline personality disorder |
| ☐ Depressive disorders | ☐ Dissociative disorders | ☐ Cluster C (anxious or fearful cluster) |
| ☐ Major Depressive Disorder | ☐ Dissociation | ☐ Avoidant personality disorder |
| ☐ Persistent Depressive Disorder | ☐ Dissociative amnesia | ☐ Dependent personality disorder |
| ☐ Bipolar disorders | ☐ Dissociative identity disorder | ☐ Obsessive-compulsive personality disorder |
| ☐ Cycling | | |

5.5 - Treatment of Psychological Disorders

- | | | |
|--|---|--|
| ☐ Evidence-based interventions | ☐ Combating maladaptive thinking | ☐ Group therapy |
| ☐ Cultural humility | ☐ Cognitive triad | ☐ Hypnosis |
| ☐ Therapeutic alliance | ☐ Applied behavior analysis | ☐ Psychoactive medication |
| ☐ Psychotropic medication | ☐ Exposure therapies | ☐ Antidepressants |
| ☐ Nonmaleficence | ☐ Systematic desensitization | ☐ Antianxiety drugs |
| ☐ Fidelity | ☐ Aversion therapies | ☐ Lithium |
| ☐ Integrity | ☐ Token economies | ☐ Antipsychotic medications |
| ☐ Respect for people's rights and dignity | ☐ Biofeedback | ☐ Tardive dyskinesia |
| ☐ Psychodynamic therapies | ☐ Cognitive-behavioral therapies | ☐ Psychosurgery |
| ☐ Free association | ☐ Dialectical behavior therapy | ☐ Lesioning |
| ☐ Dream interpretation | ☐ Rational emotive behavior therapy (REBT) | ☐ TMS (transcranial magnetic stimulation) |
| ☐ Cognitive therapies | ☐ Person-centered therapy | ☐ Electroconvulsive therapy |
| ☐ Cognitive restructuring | ☐ Unconditional positive regard | ☐ Lobotomy |
| ☐ Fear hierarchies | ☐ Active listening | |

5.1 - Introduction to Health Psychology

- **Health psychology**

- A subfield of psychology focused on promoting health and preventing/treating illness, including **psychological responses** to being sick.
- 研究人生病时的心理反应、应对方式，以及如何改善医疗系统

- **Stress**

- A state of **physiological and psychological tension** caused by stressors. Chronic stress weakens the immune system.
- 期末考试周感到身心俱疲、睡不好觉

- **Hypertension**

- **High blood pressure**, a common physical consequence of chronic stress.
- 长期高压工作导致血压升高

- **Immune suppression**

- Chronic stress **weakens the immune system**, making the body more vulnerable to illness.
- 考试周压力太大反而感冒了

- **Stressors**

- Events or conditions in the environment that **trigger** the stress response.
- 考试、失恋、经济困难、噪音

- **Eustress (motivating)**

- **Positive stress** from beneficial events that provides motivation and enhances performance.
- 比赛前的兴奋紧张感让你发挥更好

- **Distress (debilitating)**

- **Negative stress** that impairs motivation, performance, and causes emotional/physical harm.
- 亲人去世、失业带来的持续痛苦

- **Adverse childhood experiences (ACEs)**

- Potentially **traumatic events occurring before age 18** that can have lasting negative effects.
- 童年时期经历家暴、父母酗酒、被忽视

- **General adaptation syndrome (GAS)**

- **Hans Selye** 提出的三阶段压力反应模型：**Alarm** → **Resistance** → **Exhaustion**，描述身体在持续压力下如何逐步耗竭。
- 刚接到坏消息时心跳加速（警觉）→ 几周后似乎适应了（抵抗）→ 几个月后身体垮了（衰竭）

- **Alarm reaction phase**
 - The **initial reaction** to a stressor; the body activates the fight-or-flight response, releasing adrenaline and cortisol.
 - 突然被老师点名，心跳猛地加速、手心出汗
- **Resistance phase**
 - The body attempts to **adapt and cope** with the ongoing stressor; appears to be managing but resources are being consumed.
 - 连续加班几周，表面上还撑得住，但身体已经在消耗储备
- **Fight-flight-freeze response**
 - Activation of the **sympathetic nervous system**: heart rate increases, adrenaline surges, preparing the body to fight, flee, or freeze.
 - 走夜路遇到可疑的人，瞬间心跳加速、腿发软跑不动（freeze）
- **Exhaustion phase**
 - Body's resources are **depleted** after prolonged stress, leading to weakened immunity, burnout, and vulnerability to illness.
 - 长期高压工作后免疫力崩溃，反复生病、极度疲劳
- **Tend-and-befriend theory**
 - A stress response (especially in women) involving **nurturing others (tend)** and **seeking social support (befriend)**, rather than fighting or fleeing.
 - 地震后母亲先安抚孩子，然后联系亲友互相支持
- **Problem-focused coping**
 - Coping by **directly addressing the stressor** — changing the situation or one's interaction with it.
 - 考试压力大 → 制定详细的复习计划来解决问题
- **Emotion-focused coping**
 - Coping by **managing emotional reactions** when the stressor itself cannot be changed.
 - 亲人去世无法改变 → 通过冥想、深呼吸、找朋友倾诉来缓解悲伤

5.2 - Positive Psychology

- **Positive psychology**

- The scientific study of **optimal human functioning**, advocated by **Martin Seligman**. Focuses on strengths, virtues, and what helps people thrive.
- 不只研究心理疾病，而是研究什么让人活得好、活得有意义

- **Well-being**

- A broad concept encompassing **physical, emotional, and social health** — an overall sense of flourishing.
- 身体健康、情绪稳定、人际关系良好的综合状态

- **Resilience**

- The ability to **recover and bounce back** from adversity, trauma, or stress.
- 创业失败后重新振作，从头再来

- **Positive emotions**

- Emotions such as joy, gratitude, and hope that **broaden thinking** and build psychological resources.
- 帮助别人后感到的满足和快乐

- **Gratitude**

- A positive emotion emphasized in PP; research shows regularly expressing gratitude **significantly improves long-term well-being**.
- 每天写三件感恩的事，长期坚持后幸福感明显提升

- **Positive subjective experiences**

- **Internal, self-reported** experiences of happiness, satisfaction, and fulfillment.
- 回忆一次旅行时内心涌起的幸福感

- **Subjective well-being**

- A person's **self-evaluation** of their life satisfaction and happiness — how happy *they feel* they are.
- 收入不高但觉得自己生活很幸福，主观幸福感就高

- **Signature strengths**

- The **core character strengths** most central to a person's identity, organized under six virtues. Using them boosts well-being.
- 一个人最突出的特质可能是好奇心和创造力

- **Virtues**

- Six broad categories under which signature strengths are organized: **Wisdom, Courage, Humanity, Justice, Temperance, Transcendence**.
- 智慧（好奇心）、勇气（毅力）、人道（善良）、正义（公平）、节制（自控）、超越（感恩、希望）

- **Positive objective experiences**
 - **External, measurable** indicators of well-being, such as income, education, health status.
 - 稳定的收入、良好的健康指标、受教育程度
- **Happiness**
 - A state of well-being characterized by positive emotions and life satisfaction; influenced by both **genetics and circumstances**.
 - 做自己喜欢的事时感到的愉悦和满足
- **Categories of virtues**
 - The six categories: **Wisdom, Courage, Humanity, Justice, Temperance, Transcendence** — each containing specific character strengths.
 - 同上 Virtues
- **Posttraumatic growth**
 - **Positive psychological change** that emerges after struggling with a major crisis or trauma — deeper understanding of life, stronger relationships, increased personal strength.
 - 经历重病后更珍惜生活，人际关系变得更深刻，内心更强大

5.3 - Explaining and Classifying Psychological Disorders

- **Dysfunction**
 - One of the **Four Ds**: inability to **maintain daily functioning** — self-care, social relationships, or work.
 - 抑郁到无法起床上学、无法维持基本社交
- **Distress**
 - One of the Four Ds: the individual experiences **subjective suffering** from their own thoughts, emotions, or behaviors.
 - 强迫症患者因反复洗手感到极度痛苦
- **Deviation from the social norm**
 - One of the Four Ds (**Deviance**): behavior that significantly departs from cultural/social norms, with **statistical infrequency**. 但罕见≠障碍（天才也罕见）。
 - 在公共场合对着空气大声说话
- **Cultural/societal norms**
 - Standards of acceptable behavior that **vary across cultures and time periods**; what is "abnormal" is partly defined by these norms.
 - 同性恋曾被DSM列为障碍，1980年被移除——说明诊断标准随社会价值观变化
- **Stigma**
 - **Negative social attitudes** toward mental illness, leading to stereotypes, prejudice, and discrimination.
 - 别人知道你看心理医生后对你另眼相看、疏远你
- **Racism**
 - Prejudice and discrimination based on **race/ethnicity**.
 - 因种族背景在求职中被歧视
- **Sexism**
 - Prejudice and discrimination based on **sex/gender**.
 - 女性在职场中因性别被低估能力
- **Ageism**
 - Prejudice and discrimination based on **age**.
 - 认为老年人学不会新技术
- **Discrimination**
 - **Behavioral** component of prejudice — unfair treatment of individuals based on group membership.
 - 因心理疾病诊断标签而被拒绝录用

- **Diagnostic and Statistical Manual of Mental Disorders (DSM)**
 - Published by the **APA** (American Psychiatric Association). Current version is **DSM-5** (2013), containing **541 diagnoses**. Serves as the standard classification system in the US and basis for insurance.
 - 1952年首版仅128种诊断；1980年删除同性恋条目
- **International Classification of Mental Disorders (ICD)**
 - Published by the **WHO**; a globally used classification system for clinical descriptions and diagnostic codes.
 - 全球通用的疾病分类标准，不限于心理障碍
- **Eclectic approach**
 - Rejecting single-factor explanations; **integrating multiple perspectives** (biological, psychological, social) to understand and treat disorders.
 - 治疗抑郁症时同时考虑药物、认知疗法和社会支持
- **Behavioral perspective**
 - Psychological disorders result from **maladaptive learned associations** through classical and operant conditioning.
 - 被狗咬过后对所有狗产生恐惧（经典条件反射）
- **Maladaptive learned associations**
 - Unhealthy **stimulus-response connections** formed through learning that maintain problematic behavior.
 - 每次焦虑就喝酒缓解，形成了不健康的应对联结
- **Humanistic perspective**
 - Disorders arise when **self-actualization is blocked** or growth environment is deficient; emphasizes unconditional positive regard.
 - 一个人因长期被否定而无法发挥潜能，需要无条件的接纳和支持
- **Cognitive perspective**
 - Disorders stem from **maladaptive thinking patterns** and cognitive distortions; treatment involves identifying and correcting negative automatic thoughts.
 - 一次考试没考好就认为自己什么都做不好（以偏概全的认知扭曲）
- **Psychodynamic perspective**
 - Disorders originate from **unconscious conflicts and childhood experiences**; treatment involves uncovering and processing these.
 - 童年被忽视的经历导致成年后的信任问题

- **Evolutionary perspective**

- Certain psychological traits or disorders may have been **adaptive in ancestral environments**.
- 焦虑反应在远古时期帮助人类警觉危险、存活下来

- **Sociocultural perspective**

- Disorders arise from conflicts between the individual and **social/cultural norms and environments**.
- 移民因文化冲突产生适应障碍

- **Biological perspective**

- Disorders result from **neurochemical imbalances, genetic factors, or brain structure abnormalities**; treated with medication, stimulation, or surgery.
- 抑郁症与血清素水平低有关，用药物调节神经递质

- **Biopsychosocial model**

- An **integrative model**: disorders result from the interaction of **biological × psychological × social** factors. The mainstream approach in modern clinical practice.
- 抑郁症 = 基因易感性（bio）+ 消极思维模式（psych）+ 社会孤立（social）

- **Diathesis-stress model**

- Disorders are triggered when **innate genetic vulnerability (diathesis/nature)** interacts with **environmental stressors (stress/nurture)**. Having a family history ≠ inevitably developing the disorder.
- 有精神分裂症家族史（素质），但只有在经历重大压力事件后才发病（压力触发）

5.4 - Selection of Categories of Psychological Disorders

- **Neurodevelopmental disorders**

- A group of brain dysfunctions present at **birth or early childhood**, affecting behavior, memory, attention, and/or learning. Causes include genetic, physiological, and environmental factors.
- ADHD和ASD都属于这一类

- **Attention-deficit/hyperactivity disorder (ADHD)**

- Characterized by persistent patterns of **inattention** and/or **hyperactivity-impulsivity** that interfere with functioning. Requires ≥ 6 symptoms for ≥ 6 months, onset **before age 12**, across multiple settings.
- 上课无法集中注意力，坐不住，经常打断别人说话

- **Autism Spectrum Disorder (ASD)**

- Characterized by deficits in **social-emotional reciprocity, nonverbal communication, and relationship maintenance**, plus **restricted/repetitive** behaviors, interests, or activities. Onset in early childhood.
- 不理解社交暗示，对特定话题有极度强烈的兴趣，重复同样的动作

- **Schizophrenic Spectrum Disorders**

- A **psychotic disorder** involving deterioration in functioning due to abnormal perceptions, bizarre thinking, emotional dysregulation, and motor abnormalities. 不是人格分裂（split mind 是心智的分裂）。Diagnosis requires ≥ 2 symptoms for ≥ 1 month, with ≥ 5 months of impairment.
- 一个人出现幻觉、妄想，无法正常工作 and 社交

- **Positive symptoms**

- Symptoms that represent an **excess or addition** to normal functioning — things present that shouldn't be.
- 妄想、幻觉、言语紊乱、紧张症

- **Delusions**

- **False beliefs** maintained despite contradictory evidence.
- 坚信自己被监视，即使没有任何证据

- **Delusions of persecution**

- Belief that one is being **persecuted, watched, or conspired against** by others. 最常见的妄想类型。
- 坚信有人在食物里下毒

- **Delusions of grandeur**

- Belief that one is an **extraordinarily powerful or important** person with special abilities.
- 相信自己是某个王室的继承人

- **Hallucinations**
 - **False sensory perceptions** that feel vivid and real; content is usually negative. **Auditory hallucinations** are the most common type. Research shows the brain actually generates neural signals that produce the sounds.
 - 听到有声音命令自己做事，但周围没有人
- **Disorganized thinking**
 - Disrupted thought processes reflected in incoherent or illogical patterns of reasoning.
 - 无法按逻辑组织想法，思维跳跃混乱
- **Disorganized speech**
 - Incoherent or meaningless speech patterns, such as jumping between unrelated topics or inappropriate word use.
 - 说话完全没有逻辑，听的人无法理解
- **Word salad**
 - Completely **scrambled, incoherent speech** — an extreme form of disorganized speech.
 - 苹果飞机紫色跳舞.....完全无法理解的词语堆砌
- **Disorganized motor behavior**
 - Excessive, **purposeless motor activity** including agitation, bizarre postures, pacing, repetitive movements, or even aggression.
 - 反复踱步、模仿别人的动作、摆出怪异姿势
- **Catatonia**
 - As a positive symptom: excessive purposeless movement. As a negative symptom: **lack of movement or responsiveness** (catatonic stupor).
 - 阳性：不停踱步、做重复动作；阴性：保持一个诡异的姿势一动不动
- **Negative symptoms**
 - Symptoms that represent a **reduction or absence** of normal functioning. 可能被误认为抑郁或懒惰。
 - 情感平淡、缺乏动力、社交退缩
- **Flat affect**
 - Severely **diminished emotional expression**: unchanging facial expression, poor eye contact, monotone voice, reduced gestures.
 - 听到好消息或坏消息时表情完全没有变化，说话语调平平
- **Catatonic stupor**
 - A negative symptom: **lack of movement or responsiveness** to the environment; may hold unusual postures and resist being moved.
 - 像雕塑一样保持一个姿势几个小时不动，不回应任何人

- **Dopamine hypothesis**
 - The theory that **highly fluctuating dopamine levels** contribute to schizophrenia symptoms. Dopamine is the key neurotransmitter associated with schizophrenia.
 - 多巴胺水平异常波动导致幻觉和妄想等症状
- **Depressive disorders**
 - A category of mood disorders characterized by persistent **low mood, loss of motivation**, and impaired functioning.
 - 持续的悲伤、对事物失去兴趣
- **Major Depressive Disorder**
 - **Unipolar** depression: intense depressive mood, decreased interest/pleasure, energy loss, and decision-making problems for **at least 2 weeks**. Requires depressed mood or loss of interest plus ≥3-4 additional symptoms.
 - 连续两周以上每天都感到极度悲伤，无法上学上班，反复想到死亡
- **Persistent Depressive Disorder**
 - A chronic, **longer-lasting** form of depression lasting **at least 2 years**, with symptoms such as appetite changes, sleep disturbances, low energy, and negative self-perception.
 - 两年多来大部分时间都情绪低落，虽然没有重度抑郁那么严重，但一直好不了
- **Bipolar disorders**
 - Mood disorders involving **alternating episodes** between depression and mania.
 - 情绪在极度亢奋和极度低落之间来回波动
- **Cycling**
 - The **switching between manic and depressive states**; duration and intensity can vary.
 - 这个月躁狂、下个月抑郁，反复循环
- **Mania**
 - Elevated mood, **increased energy**, decreased need for sleep, racing thoughts, impulsive/risky behavior, inflated self-esteem, more talkative and sociable.
 - 连续几天不睡觉但精力充沛，疯狂消费，觉得自己无所不能
- **Depression**
 - Persistent sadness, fatigue, insomnia or hypersomnia, difficulty concentrating, feelings of worthlessness, **withdrawal** from activities and people, suicidal thoughts.
 - 躁狂期过后陷入极度疲惫和绝望，不想见任何人
- **Bipolar I Disorder**
 - Characterized by at least one full **manic episode**; may have hypomanic or major depressive episodes. **More severe** than Bipolar II.
 - 经历过完整的躁狂发作，可能住院治疗

- **Bipolar II Disorder**

- Characterized by at least one **hypomanic episode** and one major depressive episode; **never** a full manic episode. Less severe than Bipolar I.
- 有过轻躁狂（情绪高涨但没有严重到影响功能）和重度抑郁发作

- **Anxiety disorders**

- The **most common** psychological disorders in the US (19 million adults). Pathological anxiety is disruptive, irrational, and uncontrollable.
- 焦虑本身正常，但病理性焦虑是过度的、无法控制的

- **Generalized anxiety disorder (GAD)**

- Excessive anxiety about **almost everything** for ≥ 6 months, with ≥ 3 of: edginess, fatigue, poor concentration, irritability, muscle tension, sleep problems. **Free-floating anxiety** — no specific trigger.
- 什么都担心，工作、健康、家人，但说不清楚到底怕什么

- **Panic disorder**

- Characterized by recurrent, **unpredictable panic attacks** with severe physical symptoms (chest pain, choking, sweating). Lasting minutes to hours, no specific trigger. Requires ≥ 1 month of worry about recurrence or maladaptive behavioral changes.
- 突然心跳加速、胸痛、感觉快要窒息，以为自己心脏病发作

- **Panic attack**

- An episode of **intense, acute anxiety** with severe physical symptoms such as chest pain, sweating, choking, and muscle tightness.
- 突然感到极度恐惧，全身发抖、呼吸困难，持续几分钟

- **Ataque de nervios**

- A **culture-bound syndrome** common in **Latin American and Mediterranean** cultures, characterized by emotional outbursts — screaming, crying, trembling, or fainting — as a response to intense emotional events.
- 在拉丁文化中，亲人去世后出现尖叫、哭泣、颤抖甚至昏厥的情绪爆发

- **Specific phobia**

- An intense, **irrational fear** of a specific stimulus, lasting ≥ 6 months, causing immediate fear upon exposure and avoidance behavior.
- 看到蜘蛛就尖叫逃跑，明知过度但无法控制

- **Acrophobia**

- Fear of **heights**.
- 站在高楼阳台上就双腿发软、头晕

- **Arachnophobia**
 - Fear of **spiders**.
 - 看到蜘蛛就极度恐慌
- **Agoraphobia**
 - Fear of **open or public spaces** where escape might be difficult.
 - 害怕去商场、广场等开阔公共场所
- **Social anxiety disorder**
 - Intense fear and anxiety in **social situations** where one might be scrutinized; fear of being negatively evaluated. Lasts ≥ 6 months, disproportionate to actual threat.
 - 在课堂上被叫起来回答问题时极度恐惧，害怕被同学笑话
- **Taijin kyofusho**
 - A **culture-bound syndrome** from **Japan**: intense fear of **offending or embarrassing others** with one's appearance or behavior (contrast with Western social anxiety which focuses on fear of being judged oneself).
 - 日本文化中，极度害怕自己的存在让别人感到不舒服
- **Obsessive-compulsive and related disorders**
 - A category of disorders involving **obsessions, compulsions, or related repetitive behaviors** that create distress and interfere with daily functioning. Includes conditions such as OCD and hoarding disorder.
 - 一类以强迫观念、强迫行为或相关重复行为为核心的障碍，会带来痛苦并影响日常生活，包括强迫症和囤积障碍
- **Obsessive-compulsive disorder (OCD)**
 - A disorder involving both **obsessions** (intrusive, unwanted thoughts) and **compulsions** (ritualistic behaviors performed to reduce anxiety). Time-consuming and causes significant distress.
 - 反复想到门没锁好（强迫观念），于是回去检查五六次（强迫行为）
- **Obsessions**
 - Persistent, intrusive, **unwanted thoughts** the individual cannot get rid of: contamination fears, harm fears, need for symmetry, intrusive violent/sexual thoughts.
 - 脑海里不停出现自己会伤害家人的画面，越想停越停不下来
- **Compulsions**
 - **Repetitive ritualistic behaviors** performed to reduce the anxiety caused by obsessions: excessive washing, checking, arranging, counting, hoarding.
 - 每天洗手几十次直到手都裂了

- **Hoarding disorder**
 - Persistent difficulty **discarding possessions** due to a perceived need to save them, causing severe clutter and significant distress or impairment.
 - 家里堆满了旧报纸和塑料袋，完全无法丢弃，连走路的空间都没有
- **Dissociative disorders**
 - Characterized by **disruptions in consciousness, memory, identity, or perception**, causing significant impairment. Dissociation itself can be normal (e.g., highway hypnosis).
 - 开车时进入自动驾驶状态，到达目的地后完全不记得路上发生了什么
- **Dissociation**
 - A **disconnection** from one's thoughts, feelings, surroundings, or identity; can range from normal (daydreaming) to pathological.
 - 压力极大时感觉自己灵魂出窍，像在看别人的生活
- **Dissociative amnesia**
 - Inability to recall **important autobiographical information**, usually of a traumatic nature — beyond ordinary forgetting.
 - 经历车祸后完全记不起事故发生的过程
- **Dissociative identity disorder (DID)**
 - Presence of **≥2 distinct personality states**, each with its own way of perceiving and interacting with the world, causing memory gaps. Often associated with severe childhood physical/sexual abuse.
 - 一个人有多个不同的人格，彼此之间可能互不知道对方的存在
- **Trauma and stressor-related disorders**
 - Disorders involving psychological distress after exposure to traumatic/stressful events, featuring hypervigilance, severe anxiety, flashbacks, insomnia, emotional detachment, and hostility.
 - 经历战争或灾难后出现一系列心理症状
- **Hypervigilance**
 - A state of **heightened alertness** and scanning for threats.
 - 退伍军人听到巨响就立刻趴下寻找掩护
- **Flashbacks**
 - **Intrusive, vivid re-experiencing** of traumatic events as if they are happening again.
 - 突然感觉自己又回到了事故现场，画面和声音都栩栩如生
- **Insomnia**
 - Difficulty **falling or staying asleep**, commonly associated with trauma-related disorders.
 - 经历创伤后每晚都难以入睡，反复惊醒

- **Emotional detachment**
 - A state of **emotional numbness** or disconnection from one's feelings and from others.
 - 经历创伤后对任何事都感觉麻木，无法感受到快乐或悲伤
- **Hostility**
 - Increased **irritability and aggression** as a symptom of trauma-related disorders.
 - 创伤后变得暴躁易怒，经常对家人发脾气
- **Posttraumatic stress disorder (PTSD)**
 - Persistent psychological distress after trauma exposure, characterized by **intrusive memories/flashbacks, avoidance of reminders, negative cognitive/emotional changes, and heightened reactivity**. Symptoms last **>1 month** and cause significant impairment. Onset can be delayed months or years.
 - 士兵从战场回来后反复做噩梦，回避任何与战争相关的事物，听到烟花声就恐慌
- **Feeding and eating disorders**
 - Disorders defined by **altered food consumption or absorption** — too little or too much — impairing health and psychological functioning.
 - 饮食行为严重异常，影响身体健康和心理功能
- **Anorexia nervosa**
 - Characterized by **restricted food intake**, intense fear of gaining weight, and **distorted body image**, leading to significant weight loss and health complications.
 - 已经很瘦了但照镜子觉得自己很胖，极度节食
- **Bulimia nervosa**
 - Characterized by recurrent **binge eating** followed by inappropriate compensatory behaviors (self-induced vomiting, fasting, excessive exercise) to prevent weight gain.
 - 一次性吃掉大量食物，然后跑到厕所催吐
- **Personality disorders**
 - Long-standing, **rigid, maladaptive** patterns of thinking and behavior that impair social functioning. Individuals typically lack insight. **Not diagnosed before age 18** because personality is still developing. Must deviate markedly from cultural norms.
 - 一个人长期的行为模式极端且僵化，影响人际关系，但自己意识不到问题
- **Cluster A (odd or eccentric)**
 - 关键词: **Weird** (怪异)
- **Paranoid personality disorder**
 - Pattern of **distrust and suspicion** of others' motives; accusatory; misreads others' intentions.
 - 总觉得朋友在背后说自己坏话，从不向任何人吐露心事

- **Schizoid personality disorder**
 - Persistent **avoidance of social relationships** and very little emotional expression; genuinely prefers being alone; appears **cold/aloof**. More common in males.
 - 真心喜欢独处，对交朋友和谈恋爱毫无兴趣
- **Schizotypal personality disorder**
 - **Bizarre thinking, perceptions, and eccentric behavior**; distorted thinking; avoids close relationships.
 - 相信自己有心灵感应能力，穿着打扮非常怪异，让人感觉很奇怪
- **Cluster B (dramatic, emotional, or erratic)**
 - 关键词: **Wild** (狂野)
- **Antisocial personality disorder (APD)**
 - Pattern of **disregarding and violating others' rights**; manipulative, exploitative, irresponsible; no guilt. Closely linked to criminal behavior. More common in males.
 - 反复欺骗和利用他人，完全没有愧疚感，有犯罪记录
- **Borderline personality disorder**
 - **Instability** in relationships, self-image, and emotions, plus **impulsivity**; unpredictable; suicidal threats common. More common in females.
 - 对朋友的态度一天之内从极度依赖变成极度厌恶，经常用自伤威胁别人
- **Histrionic personality disorder**
 - Pattern of **excessive emotionality** (dramatic) and **attention-seeking**; self-centered; exaggerates. More common in females.
 - 说话夸张戏剧化，总要成为全场焦点，稍微被忽视就受不了
- **Narcissistic personality disorder**
 - Pattern of **grandiosity, need for admiration, and lack of empathy**; cannot accept criticism; expects special treatment. More common in males.
 - 觉得自己比所有人都优秀，要求特殊对待，完全不在乎别人的感受
- **Cluster C (anxious or fearful)**
 - 关键词: **Worried** (担忧)
- **Avoidant personality disorder**
 - Persistent **discomfort in social situations**, overwhelming feelings of inadequacy, and extreme sensitivity to **rejection/negative evaluation**.
 - 很想交朋友但极度害怕被拒绝，所以干脆回避一切社交

- **Dependent personality disorder**

- Pattern of **clinging, submissive behavior**; fear of separation; constant need to be cared for; lets others make all decisions. More common in females.
- 什么事都让别人替自己决定，极度害怕被抛弃

- **Obsessive-compulsive personality disorder**

- Intense preoccupation with **orderliness, perfectionism, and control**, to the point of losing flexibility and efficiency. 注意：这是人格障碍，和OCD不同。
- 对规则和细节极度执着，一份报告改了二十遍还不满意，反而什么都完不成

5.5 - Treatment of Psychological Disorders

- **Evidence-based interventions**

- Treatment methods supported by **scientific research and empirical evidence**, ensuring effectiveness and reliability.
- 使用经过临床试验验证有效的疗法，而不是凭感觉治疗

- **Cultural humility**

- Ongoing **self-reflection** and learning about clients' cultural backgrounds to provide culturally sensitive treatment.
- 治疗师主动了解来访者的文化背景，尊重文化差异

- **Therapeutic alliance**

- The **collaborative, trusting relationship** between therapist and client — a key factor in treatment effectiveness regardless of approach.
- 来访者感到被治疗师理解和信任，愿意敞开心扉

- **Nonmaleficence**

- **Do no harm** — avoid causing physical, emotional, or psychological harm to clients.
- 治疗师不能使用可能伤害来访者的方法

- **Fidelity**

- Maintaining **trust, honesty, and commitment** in the therapeutic relationship.
- 信守与来访者的咨询契约和承诺

- **Integrity**

- Maintaining **honesty, transparency, and consistency** in professional conduct.
- 对来访者坦诚，不隐瞒治疗方案的利弊

- **Respect for people's rights and dignity**

- Protecting clients' **privacy, confidentiality, and autonomy** — including the right to pause or stop treatment.
- 来访者有权随时暂停或终止治疗

- **Psychotropic medication**

- Medications that affect **brain chemistry** to treat psychological disorders; based on the premise that many disorders involve biological factors.
- 通过影响神经递质来缓解心理障碍的症状

- **Psychodynamic therapies**

- Based on **psychoanalysis**: uncovering unconscious content to relieve psychological tension. Traditional format: 2-3 sessions/week, up to 7 years.
- 通过探索潜意识冲突和童年经历来理解当前的心理问题

- **Free association**
 - Client freely expresses whatever comes to mind — thoughts, feelings, memories — to **uncover unconscious conflicts**. **Freudian slips** (口误) are seen as revealing true thoughts.
 - 来访者躺在沙发上想到什么就说什么，治疗师从中寻找潜意识线索
- **Dream interpretation**
 - Analyzing dream content to reveal **unconscious thoughts and unresolved conflicts**.
 - 通过分析梦境中反复出现的场景来发现潜意识中的恐惧
- **Cognitive therapies**
 - Based on the idea that **thinking colors our feelings** — thoughts mediate between events and emotional reactions.
 - 改变你对事件的想法，就能改变你的情绪反应
- **Cognitive restructuring**
 - Identifying and replacing **negative, irrational beliefs** with more positive, realistic thoughts.
 - 把考试没考好就觉得自己一无是处的想法，替换为这只是一次考试，下次可以改进
- **Combating maladaptive thinking**
 - Identifying and challenging **negative, irrational thought patterns** that cause emotional distress and dysfunction.
 - 意识到自己总是灾难化思维（一件小事就觉得天要塌了），然后主动纠正
- **Cognitive triad**
 - **Beck's** concept: negative views of **self, world, and future** — the core of depression.
 - 我不行（自我）、世界很糟糕（世界）、未来没有希望（未来）
- **Fear hierarchies**
 - A ranked list of feared situations from **least to most anxiety-provoking**, used in systematic desensitization.
 - 怕蜘蛛：看蜘蛛图片（低）→ 看真蜘蛛（中）→ 触摸蜘蛛（高）
- **Applied behavior analysis**
 - The application of **learning principles** (classical and operant conditioning) to modify behavior.
 - 用行为学原理系统地改变问题行为
- **Exposure therapies**
 - Gradual, repeated exposure to feared objects/situations in a **controlled environment** to reduce anxiety. Applicable to phobias, panic disorder, social anxiety, OCD, PTSD, GAD.
 - 让恐高症患者从低楼层开始逐步适应高度

- **Systematic desensitization**
 - Combines a **fear hierarchy** with progressive **relaxation training**; based on the principle of **reciprocal inhibition** — you cannot be deeply relaxed and deeply afraid at the same time.
 - 先学会深度放松，然后在放松状态下从最轻微的恐惧场景开始逐步面对
- **Aversion therapies**
 - Pairing an **unwanted behavior with an unpleasant stimulus** to create a negative association. Can be harmful if improperly administered.
 - 让酗酒者在喝酒时同时服用引起恶心的药物，建立喝酒=恶心的联结
- **Token economies**
 - Using **tokens as rewards** for desired behaviors (self-care, medication compliance); tokens can be exchanged for privileges. Based on operant conditioning.
 - 精神病院里患者按时吃药和整理床铺就获得代币，攒够可以换零食或活动时间
- **Biofeedback**
 - Using **electronic monitoring** to give individuals real-time information about physiological states (heart rate, muscle tension), enabling them to learn **self-regulation**. Used for stress management.
 - 通过屏幕实时看到自己的心率，练习用呼吸把心率降下来
- **Cognitive-behavioral therapies (CBT)**
 - Integrates cognitive and behavioral approaches: **thoughts, behaviors, and emotions are interconnected**. Treats depression, anxiety, PTSD, OCD.
 - 同时改变消极想法和不良行为来改善情绪
- **Rational emotive behavior therapy (REBT)**
 - Developed by **Albert Ellis**; identifies and restructures **irrational beliefs**. Considered a precursor to CBT.
 - 挑战我必须让所有人喜欢我这种非理性信念
- **Dialectical behavior therapy (DBT)**
 - Combines **acceptance of self** with **changing unhealthy patterns**. Core skills: emotional regulation, distress tolerance, interpersonal effectiveness, mindfulness. Treats **BPD**, depression, anxiety, PTSD, eating disorders.
 - 接受自己当下的情绪，同时学习更健康的应对方式
- **Person-centered therapy**
 - Provides a **supportive environment** for self-discovery and growth; focuses on the **present and future** rather than the past.
 - 治疗师不给建议，而是创造安全环境让来访者自己找到答案

- **Unconditional positive regard**
 - The therapist offers **complete acceptance and support** regardless of what the client says or does.
 - 无论来访者说了什么，治疗师都不评判，给予完全的接纳
- **Active listening**
 - The therapist **fully attends to, understands, responds to, and remembers** what the client expresses, validating their feelings.
 - 治疗师认真复述来访者的话，确认理解正确，让来访者感到被听见
- **Group therapy**
 - Therapy conducted with multiple clients simultaneously. **Advantages:** shared experiences, peer support, reduced isolation, diverse perspectives. **Disadvantages:** less individual attention, discomfort sharing in front of others.
 - 几个有社交焦虑的人一起做团体治疗，互相分享经历和支持
- **Hypnosis**
 - A state of **lowered conscious resistance** where the patient is more receptive to therapeutic suggestions; enhances access to the subconscious. **Effective** for pain and anxiety. **Not effective** for extracting accurate memories or age regression.
 - 用催眠帮助患者缓解慢性疼痛，但不能用来恢复准确的记忆
- **Psychoactive medication**
 - Same as psychotropic medication — drugs that alter **brain chemistry** to treat mental disorders.
 - 通过调节神经递质来治疗心理障碍的药物
- **Antidepressants**
 - Improve mood by affecting neurotransmitters, primarily **serotonin**. **SSRI** (Prozac, Zoloft, Lexapro): block serotonin reuptake, most common. **SNRI** (Effexor): serotonin + norepinephrine. **NDRI** (Wellbutrin): norepinephrine + dopamine.
 - 医生给抑郁症患者开了Prozac（一种SSRI），帮助提升血清素水平
- **Antianxiety drugs**
 - Reduce sympathetic brain activity, promote relaxation. **Benzodiazepines** (Xanax, Valium): fast-acting but **addictive**. **Azapirones** (Buspar): non-sedating, lower dependency risk but slower onset.
 - 焦虑发作时服用Xanax可以快速镇静，但长期使用有成瘾风险
- **Lithium**
 - A **mood stabilizer** commonly used to treat **bipolar disorder**; reduces the frequency and severity of manic and depressive episodes.
 - 双相障碍患者服用锂盐来稳定情绪波动

- **Antipsychotic medications**

- Eliminate positive symptoms of schizophrenia (hallucinations, delusions) by reducing **dopamine receptor activity**. **Typical/1st gen** (Haldol): effective for positive symptoms, high motor side-effect risk. **Atypical/2nd gen** (Risperdal, Zyprexa, Abilify): effective for both positive and negative symptoms, fewer motor side effects.
- 精神分裂症患者服用抗精神病药后幻觉和妄想明显减少

- **Tardive dyskinesia**

- A potential side effect of long-term antipsychotic use: **involuntary, repetitive movements** of the face, jaw, and limbs, related to dopamine dysregulation.
- 长期服用抗精神病药后出现无法控制的面部抽动和嘴巴咀嚼动作

- **Psychosurgery**

- Surgically altering brain tissue; a **last resort** when other treatments fail. Irreversible. Performed by psychiatrists and neurosurgeons collaboratively.
- 在所有其他治疗都无效后，作为最后手段进行脑部手术

- **Lesioning**

- Making small, **targeted destructions** to specific brain areas.
- 对特定脑区进行精确的小范围损毁来改变行为

- **Lobotomy**

- Severing connections to the **prefrontal cortex**; now largely **abandoned** due to severe side effects.
- 早期切断前额叶连接的手术，导致患者变得像行尸走肉，现已基本废弃

- **Electroconvulsive therapy (ECT)**

- Sending small electrical currents through the brain to induce brief seizures; used for **severe, treatment-resistant depression**. May cause short-term memory loss.
- 对药物无效的重度抑郁患者使用电休克治疗，可能导致短暂记忆丧失

- **TMS (transcranial magnetic stimulation)**

- **Non-invasive**; places a pulsed magnetic coil over the prefrontal area to stimulate nerve cells. Treats depression with minimal side effects.
- 用磁场无创地刺激大脑前额叶区域来治疗抑郁，副作用比ECT小很多